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A systematic review and meta-analysis of the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people

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A systematic review and meta-analysis of the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people

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Abstract

Objectives

Substantially higher prevalence rates of depressive problems including depressive symptoms, dysthymia, and major depressive disorders have been documented among homeless people for decades. However, to our knowledge, there are no previous systematic reviews that measured the global prevalence estimates of depressive symptoms, dysthymia, and major depressive disorders among homeless people.

Method

Databases including PubMed, Scopus, and Web of Science were systematically searched up to February 2020 to identify relevant studies that have reported data on the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people. A random-effect meta-analysis was performed to pool the prevalence estimated from individual studies. Subgroup and sensitivity analyses were employed to to compare the prevalence across the groups as well as to identify the source of heterogeneities. The Joanna Briggs Institute's quality assessment checklist was used to measure the study quality. Cochran's Q and the I^2 test were used to assess heterogeneity between the studies.

Results

Thirty-nine publications, including 16,718 participants, were included in the final analysis. This meta-analysis demonstrated considerably higher prevalence rates of depressive symptoms 46.72% (95% CI; 37.77-55.90), dysthymia 9.67% (95% CI; 5.49-16.47), as well as major depressive disorders 25.43% (95%CI 620.32-31.31) among homeless people. Our subgroup analysis showed that the prevalence of depressive symptoms was high among younger homeless people (<25 years of age), whereas the prevalence of major depressive disorder was high among older homeless people (>50 years of age) when compared with adults (25-50 years).

Conclusion

This review showed that nearly half, one-fourth, and one-tenth of homeless people are suffering from depressive symptoms, dysthymia, and major depressive disorders, respectively, which are remarkably higher than the reported prevalence rates in the general population. The findings

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suggest the need for appropriate mental health prevention and treatment strategies for this population group.

Keywords: Homeless, depressive symptoms, dysthymia, major depressive disorders, systematic review, meta-analysis

Strength and limitations of this study

- Estimating the prevalence of the different categories of depressive problems separately (depressive symptoms, dysthymia, and major depressive disorders) and assessing the prevalence across the lifespan are the main strengths of the present systematic review and meta-analysis.
- The vast majority of the included studies were performed in high-income countries and only three studies were conducted in low and middle-income countries, which significantly affects the global representativeness of the estimates.
- We included only peer-reviewed published articles.
- Only studies published in the English language were included in this meta-anlaysis, therefore, relevant studies published in another language might have been missed. However, both the funnel plot and and Egger's regression tests provided no evidence of substantial publication bias people.

Background

Global epidemiological evidence shows that a substantial proportion of homeless people living in both high income and low-income countries are suffering from mental, neurologic, and substance use disorders [1-3]. The reported prevalence estimates of mental, neurologic, and substance use disorders among homeless people ranged from 25% to 92 % [4-7]. Research evidence indicates that the presence of mental and substance disorders among homeless people is frequently and consistently associated with a higher risk of negative outcomes including disability and mortality from alcohol and drug use, suicide, and general medical conditions [8-12].

Regarding depressive problems, in particular, scientific evidence from numerous studies has shown that depressive problems are the commonest mental health problems among homeless people [13-17]. The reported prevalence estimates of depressive problems among the homeless people ranged from 9.5 to 76.9% across the studies [13-17]. Moreover, the vast majority of

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homeless people with depressive problems have a comorbid physical disease, mental, and substance use disorders [1, 3, 18]. Epidemiological evidence suggests that comorbid condition in patients with depressive disorders are associated with more severe disability, suffering, suicide, as well as higher mortality rates from different causes when compared with non-comorbid depressive disorders [8-11, 19-23].

Historically, little attention was paid to the mental health problems of homeless people; there are limited data available especially in low and middle-income countries indicating that psychiatric disorders including depression are not recognised and treated accordingly among the homeless people [24-26]. Therefore, studies on the burden and impacts of mental health problems including depressive disorders among the homeless would help policy-makers and program managers in developing policy solution and designing appropriate and effective preventions and treatment strategies for this population group including designing appropriate and effective preventions and treatment strategies for this population group [24, 27]. A systematic review and meta-analytic studies are among the required studies, which provide the highest levels of evidence for the decision-makers, clinicians as well as researches working in the area. However, to the best of our knowledge, there are no previous systematic reviews that measured the global prevalence estimates of depressive symptoms, dysthymia, and major depressive disorders among homeless people. Hence, the aim of this study was to systematically search, assess, and synthesize the global prevalence estimates of depressive problems among homeless people and forward implications of the findings for future research as well as clinical practice.

Methods

Research design and method

This systematic review and meta-analysis was performed in accordance with the guidelines of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [28]. We used a predesigned protocol for searching, extracting data, eligibility criteria, quality appraisal, and data analysis.

Data source and selection process

We searched three reputable databases ─PubMed, Scopus, and Web of Science to identify relevant studies. The following search strategy was used to find relevant studies in PubMed: ((("depressive

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disorder"[MeSH Terms] OR ("depressive"[All Fields] AND "disorder"[All Fields]) OR "depressive disorder"[All Fields] OR "depression"[All Fields] OR "depression"[MeSH Terms]) OR ("depression"[MeSH Terms] OR "depression"[All Fields] OR ("depressive"[All Fields] AND "symptoms"[All Fields]) OR "depressive symptoms"[All Fields]) OR ("depressive disorder"[MeSH Terms] OR ("depressive"[All Fields] AND "disorder"[All Fields]) OR "depressive disorder"[All Fields]) OR ("mental disorders"[MeSH Terms] OR ("mental"[All Fields] AND "disorders"[All Fields]) OR "mental disorders"[All Fields] OR ("mental"[All Fields] AND "disorder"[All Fields]) OR "mental disorder"[All Fields]) OR ("dysthymic disorder"[MeSH Terms] OR ("dysthymic"[All Fields] AND "disorder"[All Fields]) OR "dysthymic disorder"[All Fields] OR "dysthymia"[All Fields]) OR ("mental disorders"[MeSH Terms] OR ("mental"[All Fields] AND "disorders"[All Fields]) OR "mental disorders"[All Fields] OR ("psychiatric"[All Fields] AND "disorder"[All Fields]) OR "psychiatric disorder"[All Fields]) OR ("mental disorders"[MeSH Terms] OR ("mental"[All Fields] AND "disorders"[All Fields]) OR "mental disorders"[All Fields] OR ("mental"[All Fields] AND "illness"[All Fields]) OR "mental illness"[All Fields]) OR (("mental health"[MeSH Terms] OR ("mental"[All Fields] AND "health"[All Fields]) OR "mental health"[All Fields]) AND problems[All Fields])) AND (("homeless persons"[MeSH Terms] OR ("homeless"[All Fields] AND "persons"[All Fields]) OR "homeless persons"[All Fields] OR "homeless"[All Fields]) OR ("homeless persons"[MeSH Terms] OR ("homeless"[All Fields] AND "persons"[All Fields]) OR "homeless persons"[All Fields] OR "homelessness"[All Fields]))) AND (("epidemiology"[Subheading] OR "epidemiology"[All Fields] OR "prevalence"[All Fields] OR "prevalence"[MeSH Terms]) OR magnitude[All Fields]). Scopus and Web of Science were searched using terms and words suitable for the databases. The reference lists of included studies were hand searched to identify additional articles..

Eligibility criteria and study selection

In this systematic review and meta-analysis, studies were included if they: (1) were conducted among homeless people; (2) assessed the prevalence of depressive symptoms, dysthymia, or major depressive disorders or reported data to estimate the prevalence and (3) witeen in English lanaguge. Studies were excluded if they satisfy the following criteria: (1) they were commentaries, reviews,

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case reports, and animal studies; (2) they were letters, editorials, conference papers, books, and notes.

Methods for data extraction and quality assessment

Two independent authors (AB and GA) performed data extraction from the included studies. We extracted the following data from each study: first author(s) name, country of the study, sample size, year of publication, tools used, and a number of positive cases, and the reported prevalence of depressive symptoms, dysthymia, and major depressive disorders. The Joanna Briggs institute quality assessment tool was used to assess the quality of studies included in this meta-analysis [29]. According to this scale, the scoring of each study was conducted based on the frequency scales that are answered as yes, no, not clear (when some information is missing in the study), and not applicable (the question does not apply to that study depending on the design. The total quality score for each study was calculated based on the total numbers of positive scores.

Data synthesis and analysis

All statistical analysis was conducted by using comprehensive meta-analysis software version 3. The prevalence and 95% confidence interval were calculated using a random-effect model [30]. The I² statistics was used to measure between-study heterogeneity [30]. The values of I^2 statistics such as 25, 50 and 75% represented low, medium and high heterogeneity respectively [31]**.** The ages of the participants and the quality of studies was used to determine the possible source of heterogeneity between the studies. Publication bias was assessed by using the funnel plot and Egger's regression tests. For all analyses, the P-value for statistical significance was set at 0.05.

Patient and public involvement

Patient and the public were not involved in this study.

Results

Identification of relevant studies

After database searches performed strictly according to a predesigned protocol, 3388 studies were identified through electronic database search. Further 12 studies have been added using manual searches making the total studies 3400. Of these, we excluded 2703 articles during the assessment

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of duplicate and titles, as they did not satisfy the selection criteria. Subsequently, 328 articles were excluded after the assessment of the abstract as they did not meet the inclusion criteria. Hence, a full-text of 72 articles were retained for further evaluation and 39 of these were found to be eligible for meta-analysis . (Figure 1)

Characteristics of included studies

Overviews of the key characteristics of the studies included in this systematic review and meta-analysis are shown in Table 1. A total of 39 studies [1, 3, 13-19, 32-61], with 16,718 participants, were included in the analysis. Majority of these studies (n = 21) were conducted in the USA [13-17, 19, 32-34, 39, 41, 43, 44, 46, 47, 51, 54, 57, 58], and five were conducted in Canada [3, 18, 35, 48, 50], two conducted in Australia [52, 61], two conducted in Germany [53, 55], two conducted in the UK [38, 40], two conducted in France[36, 56], one conducted in Spain [42], one conducted in Japan[45], one conducted in Iran [49], one conducted in Serbia [37], one conducted in Netherlands [59], and one conducted in Ethiopia [1]. The eligible articles were published between 1989 and 2017. The sample size for the included studies ranged from 36 participants in Germany to 2838 participants in the USA.

Regarding the specific outcomes, 21 studies assessed depressive symptoms, 19 studies assessed major depressive disorders, and 6 studies assessed dysthymia (chronic minor depressive disorders).

Depressive symptoms among homeless people were predominantly measured using the Center for Epidemiologic Studies Depression Scale-Revised (CES-D). The CES-D was used in 11 studies, Beck's Depression Inventory (BDI) in 2 studies, Patient Health Questionnaire (PHQ) in 2 studies, Depression, Anxiety, and Stress Scale (DASS), Rost–Burnam Screener for Depression (RBSD)in one study, and one study used self-reported history of depressive symptoms.

Regarding major depressive disorders, Majority (n =11) of the studies used the Diagnostic Statistical Manual of Mental disorders (DSM) to measure major depressive disorders. While five studies used Composite International Neuropsychiatric Interview (CIDI), four studies used Mini International Neuropsychiatric Interview (MINI), and International Classification of Disease (ICD) applied in one study. All studies used DSM to measure dysthyma.

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Quality of included studies

Table 2 demonstrates the risk of bias and the quality of the included articles. Twenty-eight studies utilized an adequate sample size (71.79%) to measure the prevalence of depressive problems. About one-third of the studies scored positive on the item regarding response rate (35.89%) and the remaining studies scored negative (64.11%). To measure depressive problems, the vast majority of the studies utilized a standard instrument or valid diagnostic criteria (97.43%) and almost all studies used a study-specific questionnaire or instrument specifically designed to assess depressive problems (97.43%). All studies (100%) utilized appropriate statistical analysis to assess the prevalence of depressive problems. According to the Joanna Briggs institute quality assessment checklist, the included articles had a mean quality score of 7.71 ranging from five to nine. Twenty-five studies (64.10%) were high-quality studies (scored 7.71 and above) and the remaining were fair quality studies (scored between 5 and 7.71). (Table 2).

The prevalence of depressive symptoms in homeless people (meta-analysis)

The pooled prevalence estimates of depressive symptoms in the homeless people was found to be 46.72% (95% CI; 37.77-55.90). The heterogeneity across the studies was significant for this analysis (I^2=97.57%; p <0.001) **(figure 2).**

The prevalence of major depressive disorders in homeless people (meta-analysis)

The pooled prevalence of major depressive disorders was 25.43% (95%CI 620.32-31.31). Significant heterogeneity has been found for this analysis (I^2=97.40; p <0.001) **(figure 3).**

The prevalence of persistent depressive disorders (dysthymia) disorder in homeless people

Our random-effects meta-analysis resulted in pooled prevalence estimates of dysthymia in the homeless people 9.67% (95% CI; 5.49-16.47). This analysis also resulted in a significant heterogeneity between the studies (I^2=88.15%; p <0.001) **(figure 4).**

Sensitivity analysis

Table 3 illustrates the results of the subgroup and sensitivity analysis. We conducted a sensitivity analysis by the age of the participants and the quality of the included studies in order to explore the possible source of heterogeneity in the analysis of the prevalence of depressive problems the

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homeless people. Our analysis based on the age of the participants demonstrated that the prevalence of depressive symptom was high among younger homeless people (<25 years of age), whereas the prevalence of major depressive disorder was high among older homeless people (>50 years of age) when compared with adults (25-50 years). The prevalence of depressive symptoms was 58.21% for younger participants and it was 44.76% and 47.44% for adult and older participants respectively. In these sensitivity analyses, the observed difference in the magnitude of depressive symptoms by the age of the participants was not statistically significant (P=0.261).

Similarly to depressive symptoms we found no significant difference in the prevalence of depressive disorders by the age of the participants (P=0.839). The prevalence of depressive disorder was 22.98% for younger participants and it was 25.38% and 29.89% for adult and older participants respectively.

In our sensitivity analysis based on the quality of included studies, the pooled estimated prevalence of depressive symptoms was 40.45% for high-quality studies and it was 57.03 % for fair-quality studies, although the difference observed was not statistically significant (P=0.059**).** Likewise, the prevalence of depressive disorder was 21.75% as measured for high-quality studies and it was 325.45 % for fair-quality studies, although the difference observed was statistically significant (P=0.002**)** **(Table 2).**

Publication bias

In this review, the funnel plot was symmetric and regression tests associated with the funnel plot (Egger's test) provided no evidence of potential publication bias for the prevalence of depressive symptoms (B=5.11, SE=35.78, P=0.390) as well as depressive disorders (B=0.289, SE=3.26, P=0.930). **(Figure 5& 6).**

Discussion

To our knowledge, this is the first comprehensive systematic review and meta-analysis on the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people. We found 39 studies measuring the three types of depressive problems among homeless people: depressive symptoms, major depressive disorders and dysthymia. Our analysis resulted in a considerably higher prevalence of depressive symptoms, (46.72%), major depressive disorders

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(25.43%), as well as dysthymia (9.67%) among homeless people. These prevalence rates are notably higher compared to that of the general population, suggesting depressive symptoms, major depressive disorders and dysthymia are important and global public health issues among homeless people requiring urgent attention in terms of prevention and treatments.

The review showed that the existing literature on the prevalence of depressive symptoms, major depressive disorders and dysthymia, among homeless people demonstrated a significant variation depending on the age of the participants and the reported quality of the studies. The vast majority of the included studies were performed in high-income countries (93.4%; $n = 36$) and only a few studies were performed in low and middle-income countries (7.6%; $n = 3$). , Almost all studies used standard instruments to meaure measure the prevalence of depressive symptoms, major depressive disorders and dysthymia among the homeless.

The results of our systematic review and meta-analysis suggest that nearly half of homeless people have experienced depressive symptoms. Thus, the prevalence of depressive symptoms among homeless people is 6.5 times higher than the estimates in the general population (7.2%) [62]. One of the possible explanations for this difference could be the higher prevalence rates of acute and chronic medical conditions, including HIV/AIDS, tuberculosis, and other medical problems among the homeless people that possibly increase the risks of psychiatric disorders including depressive symptoms as reported in many previous studies [63-68]. Another possible explanation is that homeless people are more likely to experience sexual assault and physical abuse that is frequently and consistently associated with a higher risk of depressive problems among the exposed individuals [69-72]. Finally, when compared with the general population the magnitude of substance use disorders, as well as other psychiatric problems, are considerably high among homeless people, which are associated with greater risks of depressive symptoms among the homeless [1, 2, 7, 54, 73, 74].

Our study showed that more than one-fourth of the homeless people had major depressive disorders, which was 2.48 times higher than the estimates from a previous systematic review conducted in Western countries among the homeless (11.4%) [75]. The reason for variation in the prevalence could be due to: (1) the previous study was conducted only in Western countries and the current study included studies from developed as well as developing countries. Thus, the potential reason for the higher in the prevalence of depressive disorders in the current study may

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be due to the inclusion of new studies including studies from low and middle-income countries. (2), the variation in characteristics of the participants across the countries in terms of experiencing the potential contributing factors that increase the risk of major depressive disorders.

Surprisingly, the prevalence of major depressive disorders in the current study was 5 times higher than the global prevalence of MDD in the general population reported by a recent meta-analysis (5%) [76]. Similarly, pooled prevalence estimates in the current study 1.5 to 3 times higher than. the pooled prevalence estimates of dysthymia in the general population (3-6%) [77]. As discussed earlier the possible factors for the higher prevalence in homeless people could be due to the presence of stressful life events, as well as comorbid medical, mental, and substance use disorders among the homeless people [1, 2, 7, 54, 63-74].

This review also showed that the prevalence of depressive symptoms was high among younger homeless people (<25 years of age), whereas the prevalence of major depressive disorder was high among older homeless people (>50 years of age) when compared with adults (25-50 years). The findings support the views that depressive disorders are common in older people than younger adults and they are more likely to be severe in presentation, display cognitive changes, and more complex presentations with high somatic symptoms [78]. The possible risk factors for these include age-related neurobiological changes, cognitive diathesis, and increased rates of stressful life events [79-81].

The implication of the findings

Our findings suggest the urgent needs of robust studies to explore the possible reasons for the higher prevalence of depressive symptoms, depressive disorders and dysthymia among homeless people as compared to the reported magnitude in the general population. In addition, further studies to estimate the prevalence of depressive problems in low and middle-income countries are warranted. Also, early screening and prevention based on coordinated public health approaches are needed to address depressive problems among the homeless.

Conclusion

This systematic review and meta-analysis revealed that the prevalence of depressive symptoms, dysthymia, and depressive disorders are notably high among homeless people, underlying the importance of early screening and preventions for depressive problems among the homeless

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people. Further strong studies are needed to examine the possible reasons for the higher prevalence of depressive problems among homeless people. Also, future robust and informative studies need to investigate better mechanisms of prevention, screening and detection management of those problems among homeless people.

Abbreviations

AIDS: Acquired Immune Deficiency Syndrome

BDI: Beck's Depression Inventory

DASS: Depression, Anxiety, and Stress Scale

CIDI: Composite International Neuropsychiatric Interview

DSM IV-TR Diagnostic Statistical Manual of Mental disorders fourth, Edition, text revised DSM-5-TR: Diagnostic Statistical Manual of Mental disorders fifth, Edition, text revised HIV: Human Immune Deficiency Virus

ICD: International Classification of Disease

MINI: Mini International Neuropsychiatric Interview

NOS: Newcastle-Ottawa Scale

PHQ: Patient Health Questionnaire

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

PSE: Present State Examination

RBSD: Rost–Burnam Screener for Depression

UK: United Kingdom

USA: United States of America

WHO: World Health Organization

Declarations

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Ethics approval and consent to participate

N/A

Consent for publication

N/A

Availability of data and material

All data generated or analyzed during this study are included in this article.

Competing interests

The authors declare that there are no competing interests.

Funding

The authors declare that there is no funding.

Authors' contributions

The author (GA) conceptualized the study, conducted analyses, write-up and approval of the final manuscript. AB was performed the search, data extraction, quality assessment, write the discussion, draft and approved the final manuscript. LT was participated in discussion and consensus and approved the final manuscript.

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Table 1. Distribution of studies on depression in homeless people included in qualitative and quantitative analysis based on year, study design, sample size, instrument, country, response rate, study population and prevalence.

Author(year)(reference number)	Sample size	Tool	Country	Study population	Outcome (magnitude of depressive problems)
Landefeld et al, 2017 [19]	348	CES-D	USA	Old adults age greater than= 50	53.5%(n=185)
Bacciardi S et al, 2018 [32]	489	MINI-6	USA	Adult with TBI	35.78%(n=175)
Rogers et al. (2017) [33]	124	RBSD	USA	Adults≥50	52(n=64)
Lee KH et.al 2017 [13]	156	CESD	USA	ADULTS	120 (77.5%)
Palar K et.al (2015) [34]	346	BDI-II	USA	Adults with HIV	139 (35.8%)
Noel F et al. (2016) [35]	497	MINI,	Canada	Adults, MENTALLI ILL	40%(n=199)
Roze et.al. (2018) [36]	733	CIDI	France	Women Adults	28.8%
Sarajlija et.al. (2014) [37]	104	BDI-II	Serbia	Adults	15.4%(n=16) MDD 62.5% (n=65/104) BDI-depressive symptoms

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Berg et al.2005 [14]	415	CES-D	USA	Adults(18-55) with latent TB	50%(n=209)
Topolovec-Vranic 2017 [18]	990	MINI 6.0	Canada	Adults with mental lines/TBI	42.1% (417/990
Coohey Y C et,=.al (2015) [16]	457	DSM-IV	USA	Adults	26.25%(n=)
Pluck, G.(2008) [38]	50	Zung Self-rating Depression Scale	UK	Adults	62.14%(n=31)
Fletcher JB et.al. (2017) [39]	131	SCID-II	USA	Adults who were have sex with men(MSM)	39%(n=51)
La Gory et.al (1990) [40]	150	CES-D	UK	Adult	75% (N=150)
Susser et.al. (1989) [41]	223	CES-D	USA	Adults(Men)	33%(n=74)
Munoz et.al (1998) [42]	262 Madrid 1563 Los Angeles	CIDI DIS	Spain	Adult	MDD Madrid 21% (55/262) MDD los Angeles 21.2 (332/1563) Dysthemia madrid 17.7 47/262) Los Angelees 14.8 (231)
Nyamathi et.al. (2012) [43]	267	CES-D	USA	Adult (18-46) with G/B	62.2%(n= 166)
Ghose T. et.al (2013) [44]	2838	ICD	USA	Adults VETRIAN	9.5%(n=269)

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Okamura T et.al (2014) [45]	423	PHQ2(two question Instrument)	JAPAN	adults	28.9%(N=119/412)
Brown et.al (2013) [15]	472	PHQ-9	USA	adults aged 50 years and older	39.8%(N=99)
Logan et.al 2013 [46]	208	Not mentioned	USA	young adults aged 18–25 years	64.4(N=134)
Rhoades et.al (2013) [47]	305	CESD	USA	Adult men heterosexual(age > 18)	46.36%(n=141)
Strehlua et.al (2012) [48]	196	MINI Version 5.0.0	CANADA	Women with age 19-25	29%(n=56)
Prinsloo et.al (2012) [49]	38	SCID-I	Iran	Adult with age ≥ 26	42.1%(n=16
Whitbeck et.al. (2012) [50]	112	UM-CIDI	Canada	ADULT	29.7%(n=33)
Hadland et.al (2011)[3]	559	CES-D	CANADA	Young with age 14-26	43.4%(n=194)
Crawford et.al (2011)[17]	241	UM-CIDI	USA	Young women with age b/n 16 and 19	32.50%(n=73)
Nyamathi et.al (212) [51]	156	CES-D	USA	Young Adults	52.60%(n=82)
Larney et.al (2009) [52]	105	DASS	Australia	Adult	44.76%(n=47)
Greifenhage et.al (1997) [53]	36	DIS/DSM-III	Germany	Adult women	MDD 50%(n=16) Dysthymia 6 %(n=2)

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Koegel et.al 1998 [54]	445	DIS/DSM-III	USA	Adult	MDD 18.3%(n=60) Dysthymia 9.3% (n=41)
Fichter et.al (2001) [55]	265	SCID-IV	Germany	Adult men	21.1%(n=59)
Kovess et.al (1999) [56]	838	CIDI	France	Adult	33.7%(n=241)
Rohde et.al. (2001) [57]	523	SCID -IV	USA	Adolescent age less than 21	Depression 17.6%(n=92) MDD 12.2% (N=61) Dysthymia 6.5% (n=34)
Spinelli et.al (2017) [58]	350	CES-D	USA	Adult with age 50 and above	38.3%(n=134)
Van Straaten et al 2014 [59]	385	DASS	Netherlands	HOMELESS	41.55(160)
Bassuk el et.al 1986 [60]	80	DSM	USA	Adult	12.5% (n=10)
Herman H et.al , 1989 [61]	382	DSM	Australia	Adult	20% (n=78)
Ayano et.al 2017 [1]	456	DSM	Ethiopia	adult	9.6% (n=44)

Key: BDI: Beck's Depression Inventory; DASS: Depression, Anxiety, and Stress Scale; CIDI: Composite International Neuropsychiatric Interview; DSM IV-TR Diagnostic Statistical Manual of Mental disorders fourth, Edition, text revised DSM-5-TR: Diagnostic Statistical Manual of Mental disorders fifth, Edition,

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text revised HIV: Human Immune Deficiency Virus; ICD: International Classification of Disease; MINI: Mini International Neuropsychiatric Interview; PHQ: Patient Health Questionnaire RBSD: Rost–Burnam Screener for Depression; UK: United Kingdom; USA: United States of America

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Table 2: Qualities of studies included in the systematic review and meta-analysis

Study name	**Response**									
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Total
Landefeld et al, 2017 [19]	Y	N	Y	Y	Y	Y	Y	Y	U	7
Bacciardi S et al, 2018 [32]	Y	N	Y	Y	Y	Y	Y	Y	U	7
Rogers et al. (2017) [33]	Y	N	N	Y	Y	N	Y	Y	U	6
Lee KH et.al 2017 [13]	Y	N	N	Y	Y	N	Y	Y	U	6
Palar K et.al (2015) [34]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Noel F et al. (2016) [35]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Roze et.al. (2018) [36]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Sarajlija et.al. (2014) [37]	Y	N	Y	Y	Y	Y	Y	Y	Y	8
Berg et al.2005 [14]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Topolovec-Vranic 2017 [18]	N	Y	Y	Y	Y	Y	Y	Y	U	7
Coohey Y C et,=.al (2015) [16]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Pluck, G.(2008) [38]	N	N	N	Y	Y	Y	Y	Y	U	5
Fletcher JB et.al. (2017) [39]	N	N	N	Y	Y	Y	Y	Y	U	5
La Gory et.al (1990) [40]	Y	Y	N	Y	Y	Y	Y	Y	U	7
Susser et.al. (1989) [41]	Y	Y	N	Y	Y	Y	Y	Y	U	7
Munoz et.al (1998) [42]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Nyamathi et.al. (2012) [43]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Ghose T. et.al (2013) [44]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Okamura T et.al (2014) [45]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Brown et.al (2013) [15]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Logan et.al 2013 [46]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Rhoades et.al (2013) [47]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Strehlua et.al (2012) [48]	N	Y	Y	Y	Y	Y	Y	Y	U	8
Prinsloo et.al (2012) [49]	N	Y	N	Y	Y	Y	Y	Y	Y	7
Whitbeck et.al. (2012) [50]	N	Y	N	Y	Y	Y	Y	Y	U	6
Hadland et.al (2011)[3]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Crawford et.al (2011)[17]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Nyamathi et.al (212) [51]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9

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Larney et.al (2009) [52]	N	Y	N	Y	Y	Y	Y	Y	Y	7
Greifenhage et.al (1997) [53]	N	Y	N	Y	Y	Y	Y	Y	Y	7
Koegel et.al 1998 [54]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Fichter et.al (2001) [55]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Kovess et.al (1999) [56]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Rohde et.al. (2001) [57]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Spinelli et.al (2017) [58]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Van Straaten et al 2014 [59]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Bassuk el et.al 1986 [60]	N	Y	N	Y	Y	Y	Y	Y	U	6
Herman H et.al , 1989 [61]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Ayano et.al 2017 [1]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9

Keys:

A. Q1-Q9 represents questions used to assess the quality of included studies, which are listed below.

Q1. Was the sample frame appropriate to address the target populations?

Q2. Were the study participants sampled in appropriate way?

Q3. Was the sample size adequate?

Q4. Were the study subjects and setting described in details?

Q5. Was the data analysis conducted with sufficient coverage of the identified sample?

Q6. Was a valid method used in the identification of conditions?

Q7. Was the condition measured in a standard, reliable way for all participants?

Q8. Was there an appropriate statistical analysis?

Q9. Was the response rate adequate, and if not, was the low response rate managed appropriately?

B. Y, yes; N, no; U, unclear; NA, not applicable

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Table 3: Subgroup and sensitivity analysis of prevalence of depressive problems among homeless participants by age and quality of the studies

Subgroup	**Number of studies**	**Estimates**		Heterogeneity across the studies		Heterogeneity between groups (*P* value
		Prevalence (%)	**95% Confidence interval**	**I^2 (%)**	**P-value**	
Age (depressive symptom)						
Adult	13	44.76	32.95-57.20	98.04	<0001	0.261
Older	3	47.44	37.22-57.88	88.14	<0001	
Younger	2	58.21	46.13-69.38	88.60	<0001	
Age (depressive disorders)						
Adult	16	25.38	19.44-32.40	97.72	<0001	0.839
Older	2	29.89	14-84-51.05	95.28	<0001	
Younger	3	22.98	11.96-39.58	95.79	<0001	
Quality of the study (depressive symptoms)						
High	11	40.45	29.68-52.23	93.30	<0.0001	0.059
Fair	7	57.03	44.39-68.81	98.17	<0.0001	
Quality of the study (depressive disorders)						
High	14	21.75	16.67-27.86	81.24	<0.0001	0.002
Fair	7	35.45	29.31-42.11	97.43	<0.0001	

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Figure legends

Figure 1: **PRISMA flowchart of review search**

This figure illustrates the process of searching for relevant studies from the three reputable databases including identification, selection, eligibility and inclusion of the studies depending on the predefined criteria**.**

Figure 2: Forest plot of the prevalence of depressive symtoms among homeless people: a meta-analysis

The figure shows the results of the meta-analysis of the studies on depressive symtoms among homeless people using random-effect model.

Figure 3: Forest plot of the prevalence of major depressive disorders among homeless people: a meta-analysis

The figure shows the results of the meta-analysis of the studies on major depressive disorders among homeless people using random-effect model.

Figure 4: Forest plot of the prevalence of dysthemia among homeless people: a meta-analysis

The figure shows the results of the meta-analysis of the studies on dysthemia among homeless people using random-effect model.

Figure 5: Publication bias for depressive symtoms among homeless people

Figure 6: Publication bias for major depressive disorders among homeless people

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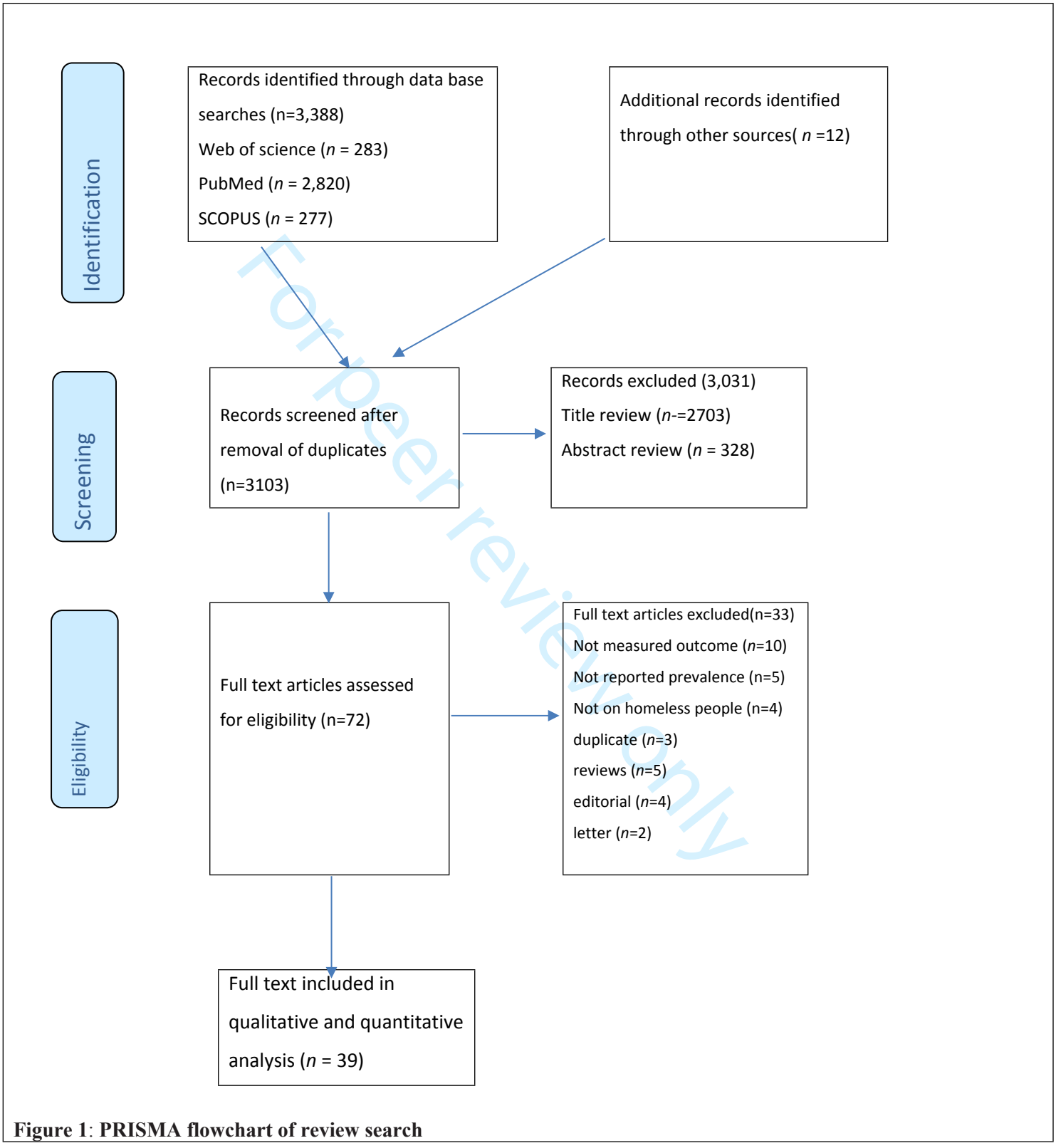
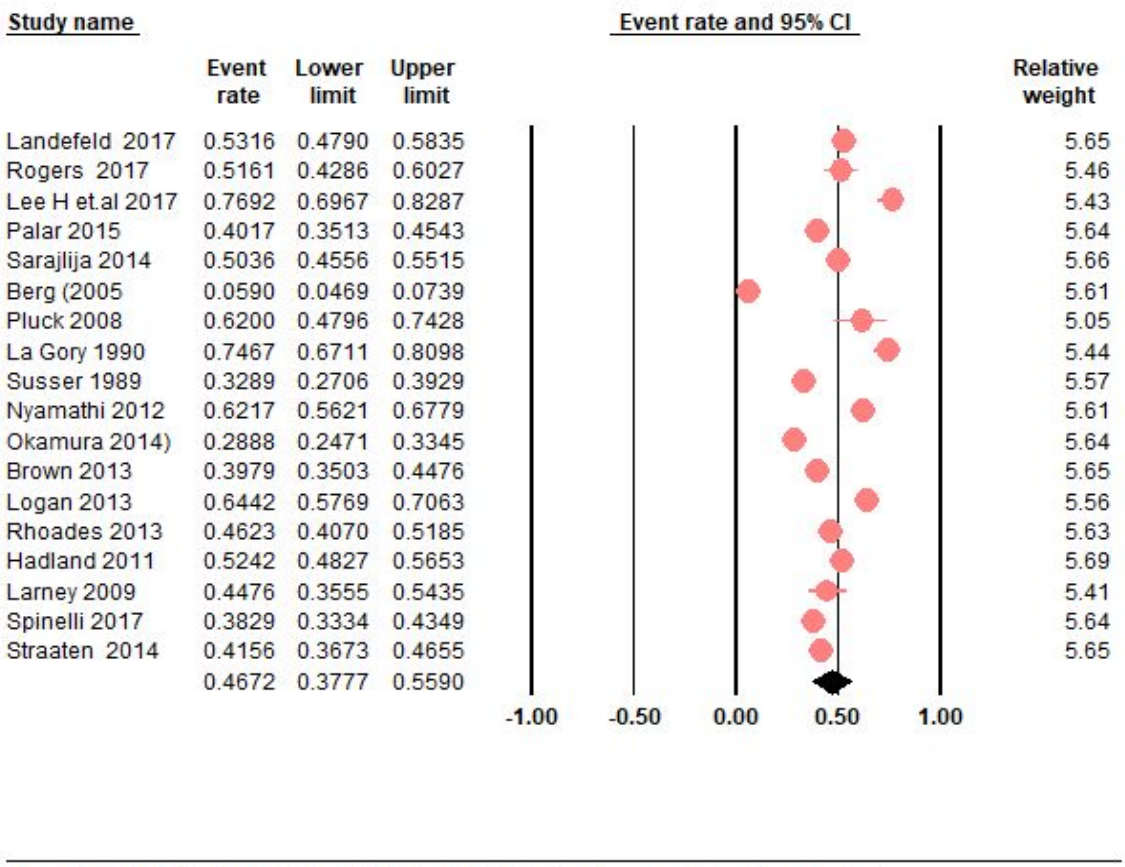


Figure 1: **PRISMA flowchart of review search**

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Figure 2: The prevalence of depressive symptoms among homeless people: a meta-analysis

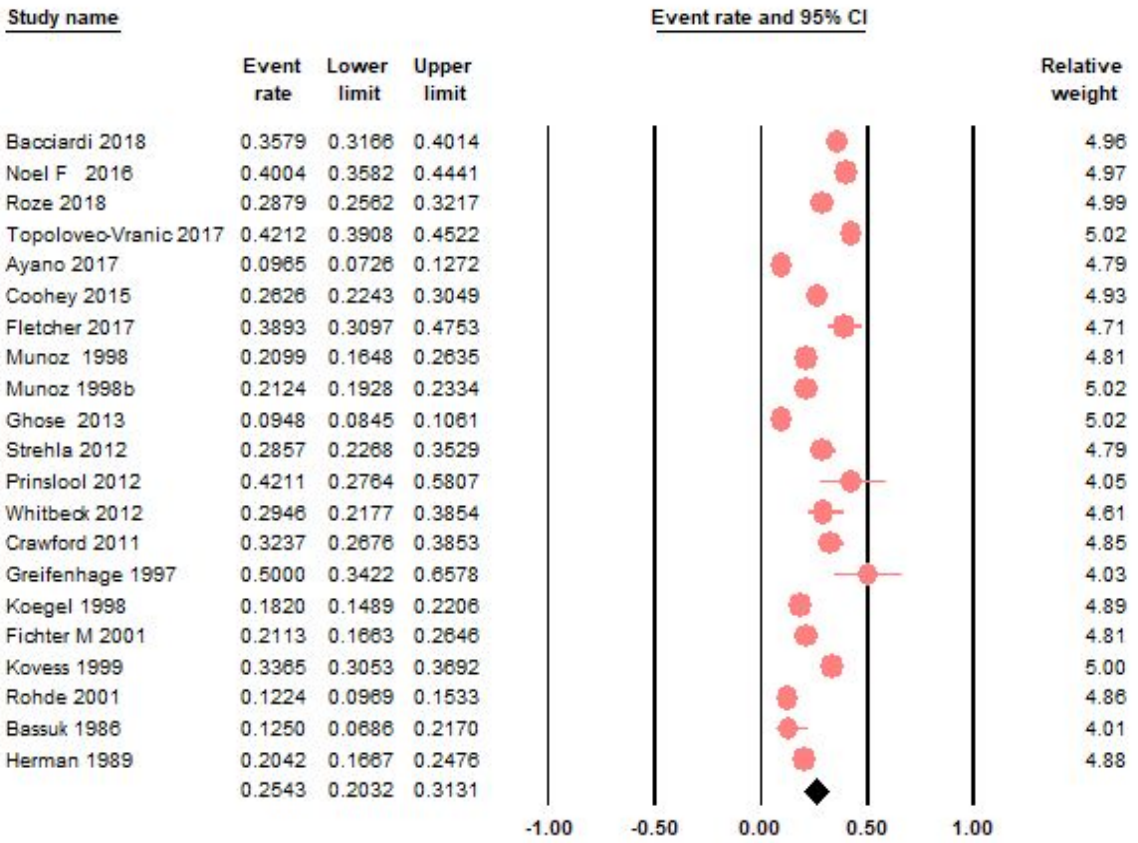


Keys: I^2=97.57%, Q=698.54; df=17; P<0.0001; Based on random effect analysis

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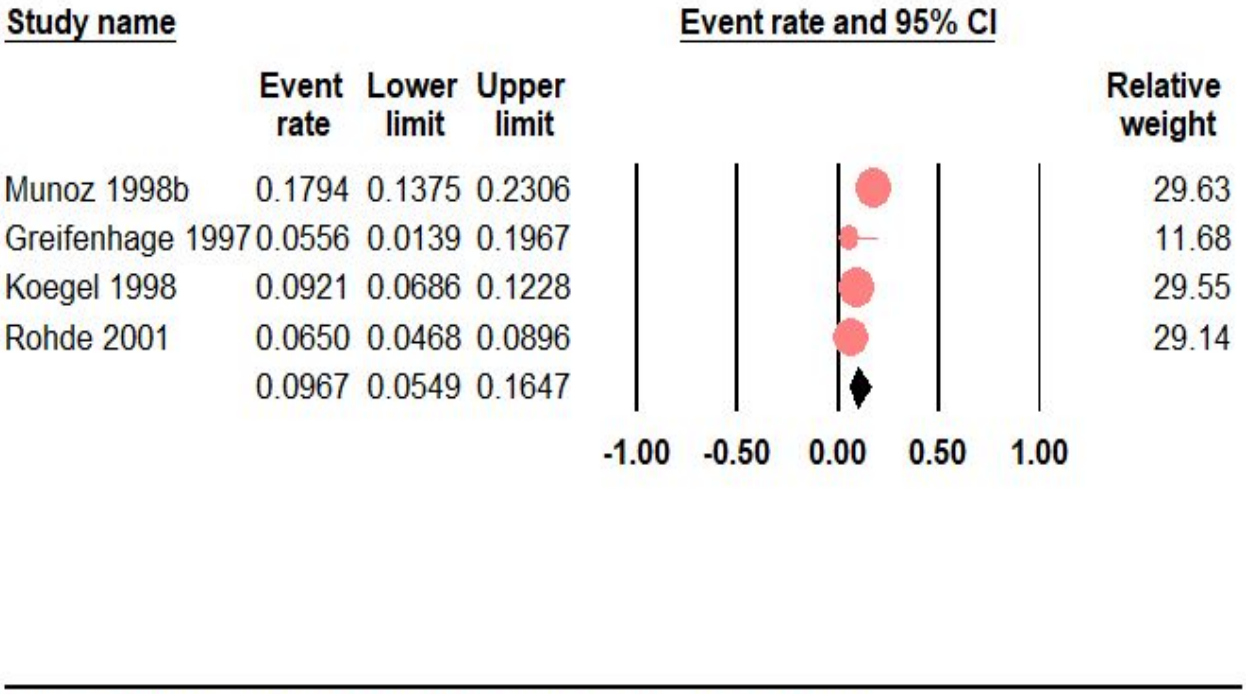
Figure 3: The prevalence of major depressive disorder among homeless people: a meta-analysis



Keys: I^2=97.40%, Q=770.43; df=20; P<0.0001; Based on random effect analysis

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Figure 4: The prevalence of dysthemia among homeless people: a meta-analysis



Keys: I^2=88.15%, Q=25.32; df=3; P<0.0001; Based on random effect analysis

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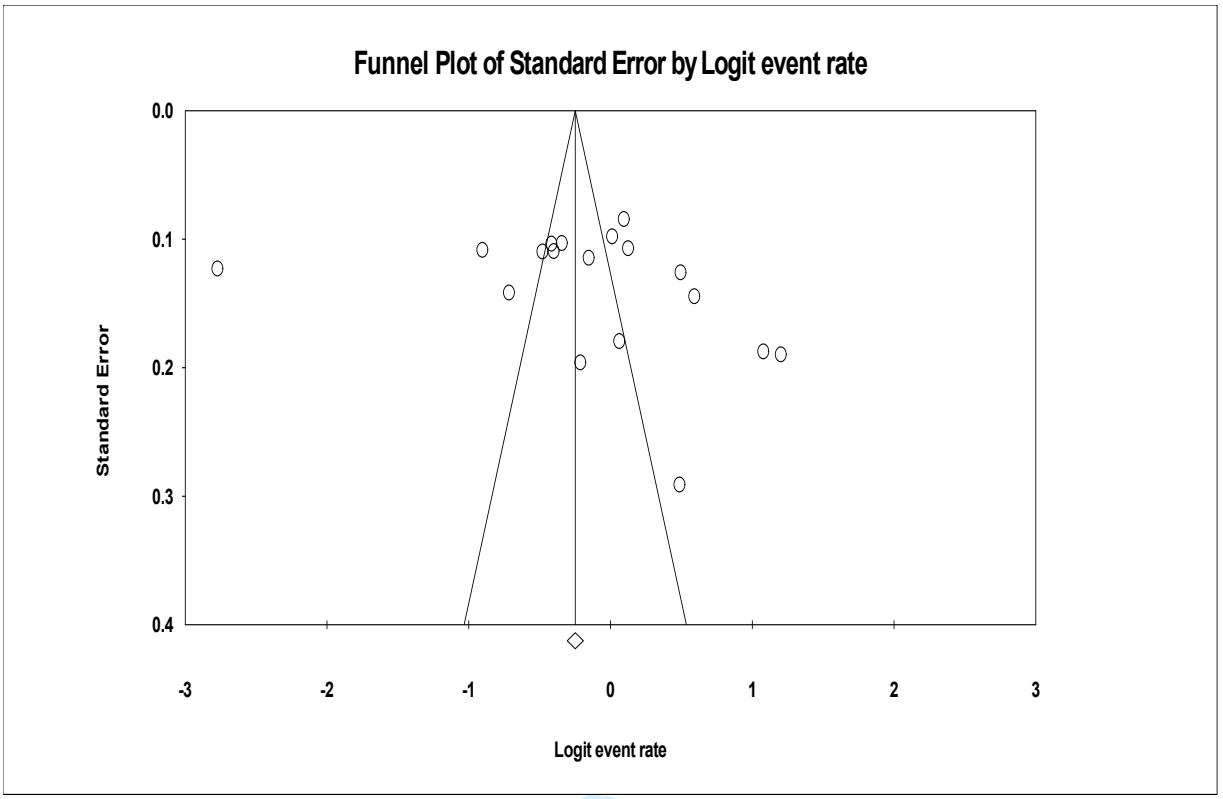


Figure 5: The risk of publication bias for the prevalence of depressive symptoms among homeless people

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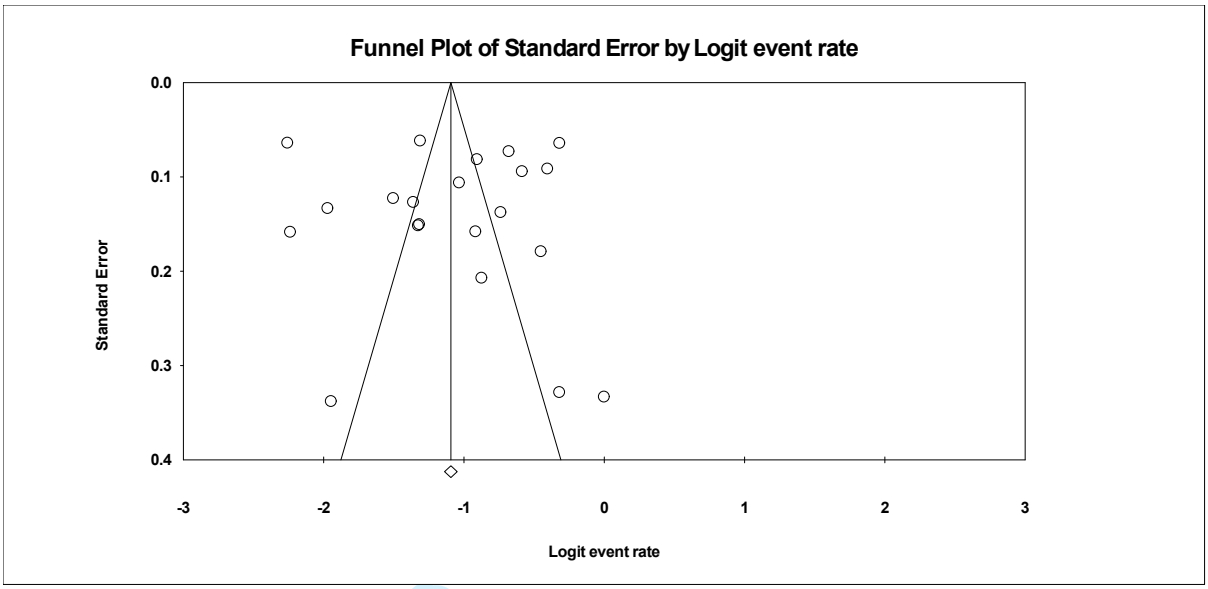


Figure 6: The risk of publication bias for the prevalence of major depressive disorders among homeless

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4-5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7-8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7-8
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7-9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9-11
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	13

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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A systematic review and meta-analysis of the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people

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A systematic review and meta-analysis of the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people

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Abstract

Objectives

To assess the global prevalence estimates of depressive symptoms, dysthymia, and major depressive disorders among homeless people.

Design

Systematic review and meta-analysis

Data sources

Databases including PubMed, Scopus, and Web of Science were systematically searched up to February 2020 to identify relevant studies that have reported data on the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people.

Eligibility criteria

Original epidemiologic studies written in English that addressed the prevalence of depressive problems among homeless people

Data extraction and synthesis

A random-effect meta-analysis was performed to pool the prevalence estimated from individual studies. Subgroup and sensitivity analyses were employed to compare the prevalence across the groups as well as to identify the source of heterogeneities. The Joanna Briggs Institute's quality assessment checklist was used to measure the study quality. Cochran's Q and the I^2 test were used to assess heterogeneity between the studies.

Results

Forty publications, including 17,215 participants, were included in the final analysis. This meta-analysis demonstrated considerably higher prevalence rates of depressive symptoms 46.72% (95% CI; 37.77-55.90), dysthymia 8.25% (95% CI; 4.79-11.86), as well as major depressive disorders 26.24% (95%CI 21.02-32.22) among homeless people. Our subgroup analysis showed that the prevalence of depressive symptoms was high among younger homeless people (<25 years of age), whereas the prevalence of major depressive disorder was high among older homeless people (>50 years of age) when compared with adults (25-50 years).

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Conclusion

This review showed that nearly half, one-fourth, and one-tenth of homeless people are suffering from depressive symptoms, dysthymia, and major depressive disorders, respectively, which are remarkably higher than the reported prevalence rates in the general population. The findings suggest the need for appropriate mental health prevention and treatment strategies for this population group.

Keywords: Homeless, depressive symptoms, dysthymia, major depressive disorders, systematic review, meta-analysis

Strength and limitations of this study

- Estimating the prevalence of the different categories of depressive problems separately (depressive symptoms, dysthymia, and major depressive disorders) and assessing the prevalence across the lifespan are the main strengths of the present systematic review and meta-analysis.
- The vast majority of the included studies were performed in high-income countries and only three studies were conducted in low and middle-income countries, which significantly affects the global representativeness of the estimates.
- We included only peer-reviewed published articles.
- Only studies published in the English language were included in this meta-analysis, therefore, relevant studies published in another language might have been missed.

Background

Global epidemiological evidence shows that a substantial proportion of homeless people living in both high income and low-income countries are suffering from mental, neurologic, and substance use disorders [1-3]. The reported prevalence estimates of mental, neurologic, and substance use disorders among homeless people ranged from 25% to 92 % [4-7]. Research evidence indicates that the presence of mental and substance disorders among homeless people is frequently and consistently associated with a higher risk of negative outcomes including disability and mortality from alcohol and drug use, suicide, and general medical conditions [8-12].

Regarding depressive problems, in particular, scientific evidence from numerous studies has shown that depressive problems are the commonest mental health problems among homeless

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people [13-17]. The reported prevalence estimates of depressive problems among the homeless people ranged from 9.5 to 76.9% across the studies [13-17]. These prevalence rates are higher than the estimates in the general population (5-7.2%) [18, 19]. Moreover, the vast majority of homeless people with depressive problems have comorbid physical disease, mental, and substance use disorders [1, 3, 20]. Epidemiological evidence suggests that comorbid conditions in patients with depressive disorders are associated with more severe disability, suffering, suicide, as well as higher mortality rates from different causes when compared with non-comorbid depressive disorders [8-11, 21-25].

However, to the best of our knowledge, there is no previous systematic review and meta-analysis that measured the global prevalence estimates of depressive symptoms, dysthymia, and major depressive disorders among homeless people. This study fills this gap in the literature by analyzing the global prevalence estimates of those depressive problems among homeless people. The finding from such analysis will provide robust data on depressive problems among homeless individuals, which in turn helps policy-makers and program managers in developing policy solutions and designing appropriate and effective intervention strategies for those population groups.

Methods

Data sources and search strategies

This systematic review and meta-analysis was performed in accordance with the guidelines of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [26]. **A** systematic search of relevant studies was carried out in three reputable databases ─PubMed, Scopus, and Web of Science using keywords and combinations tailored to each database. The details about the searching technique are in supplementary material 1. In addition, the reference lists of included studies were hand-searched to identify additional articles.

Eligibility criteria and study selection

In this systematic review and meta-analysis, studies were included if they: (1) were conducted among homeless people; (2) assessed the prevalence of depressive symptoms, dysthymia, or major depressive disorders or reported data to estimate the prevalence and (3) written in English language. Studies were excluded if they satisfy the following criteria: (1) they were commentaries,

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reviews, case reports, and animal studies; (2) they were letters, editorials, conference papers, books, and notes.

Selection of studies for the inclusion in the systematic review and meta-analysis

In this systematic review and meta-analysis, the corresponding author (GA) identified articles and consequently evaluated them by their titles and abstract based on the eligibility criteria. Full-text articles were selected for further assessment by the author. This author further appraised the full text of each study and subsequently retained those full-text articles to be included in the final analysis.

Definition of key terms and concepts

Depressive disorders and dysthymia

This systematic review and meta-analysis combined studies from a wide range of settings and countries across the globe that reported the lifetime prevalence estimate of major depressive disorders as well as dysthymia according to standard diagnostic criteria's, which were assessed by using standard diagnostic instruments such as the Diagnostic Statistical Manual of Mental disorders (DSM), Schedule for Clinical Assessment of Neuropsychiatry (SCAN), International Classification of Disease (ICD), and Composite International Neuropsychiatric Interview (CIDI) and Mini-International Neuropsychiatric Interview (MINI).

Depressive symptoms

The prevalence of depressive symptoms in this review indicates the current prevalence as measured by the screening instruments such as the Centre for Epidemiologic Surveys for Depression (CES-D), Depression, Anxiety, and Stress Scale (DASS), Patient Health Questionnaire (PHQ-9), and Zung Self-rating Depression Scale.

Methods for data extraction and quality assessment

Two independent authors (AB and GA) performed data extraction from the included studies. We extracted the following data from each study: first author(s) name, country of the study, sample size, year of publication, tools used, and a number of positive cases, and the reported prevalence of depressive symptoms, dysthymia, and major depressive disorders. The Joanna Briggs institute quality assessment tool was used to assess the quality of studies included in this meta-analysis [27]. According to this scale, the scoring of each study was conducted based on the frequency

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scales that are answered as yes, no, not clear (when some information is missing in the study), and not applicable (the question does not apply to that study depending on the design. The total quality score for each study was calculated based on the total numbers of positive scores.

Data synthesis and analysis

All statistical analysis was conducted by using comprehensive meta-analysis software version 3. The prevalence and 95% confidence interval were calculated using a random-effect model [28]. The I² statistics was used to measure between-study heterogeneity [28]. The values of I^2 statistics such as 25, 50 and 75% represented low, medium, and high heterogeneity respectively [29]**.** The ages of the participants and the quality of studies was used to determine the possible source of heterogeneity between the studies. Publication bias was assessed by using the funnel plot and Egger's regression tests. For all analyses, the P-value for statistical significance was set at 0.05.

Patient and public involvement

Patient and the public were not involved in this study.

Results

Identification of relevant studies

After database searches performed strictly according to a predesigned protocol, 3388 studies were identified through an electronic database search. Further 13 studies have been added using manual searches making the total studies 3401. Of these, we excluded 2703 articles during the assessment of duplicate and titles, as they did not satisfy the selection criteria. Subsequently, 328 articles were excluded after the assessment of the abstract as they did not meet the inclusion criteria. Hence, a full-text of 73 articles were retained for further evaluation and 40 of these were found to be eligible for meta-analysis. (Figure 1)

Characteristics of included studies

Overviews of the key characteristics of the studies included in this systematic review and meta-analysis are shown in Table 1. A total of 40 studies [1, 3, 13-17, 20, 21, 30-60], with 17,215 participants, were included in the analysis. Majority of these studies (n = 21) were conducted in the USA [13-17, 21, 30-32, 37, 39, 41, 42, 44, 45, 49, 52, 55, 56], and six were conducted in Canada [3, 20, 33, 46, 48, 60], two conducted in Australia [50, 59], two conducted in Germany

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[51, 53], two conducted in the UK [36, 38], two conducted in France[34, 54], one conducted in Spain [40], one conducted in Japan[43], one conducted in Iran [47], one conducted in Serbia [35], one conducted in Netherlands [57], and one conducted in Ethiopia [1]. The eligible articles were published between 1989 and 2017. The sample size for the included studies ranged from 36 participants in Germany to 2838 participants in the USA.

Regarding the specific outcomes, 21 studies assessed depressive symptoms, 20 studies assessed major depressive disorders, and 5 studies assessed dysthymia (chronic minor depressive disorders).

Depressive symptoms among homeless people were predominantly measured using the Center for Epidemiologic Studies Depression Scale-Revised (CES-D). The CES-D was used in 11 studies, Beck's Depression Inventory (BDI) in 2 studies, Patient Health Questionnaire (PHQ) in 2 studies, Depression, Anxiety, and Stress Scale (DASS), Rost–Burnam Screener for Depression (RBSD)in one study, and one study used a self-reported history of depressive symptoms.

Regarding major depressive disorders, the Majority (n =11) of the studies used the Diagnostic Statistical Manual of Mental disorders (DSM) to measure major depressive disorders. While five studies used Composite International Neuropsychiatric Interview (CIDI), five studies used Mini-International Neuropsychiatric Interview (MINI), and International Classification of Disease (ICD) applied in one study. All studies used DSM to measure dysthymia.

Quality of included studies

Table 2 demonstrates the risk of bias and the quality of the included articles. Twenty-eight studies utilized an adequate sample size (71.79%) to measure the prevalence of depressive problems. About one-third of the studies scored positive on the item regarding response rate (35.89%) and the remaining studies scored negative (64.11%). To measure depressive problems, the vast majority of the studies utilized a standard instrument or valid diagnostic criteria (97.43%), and almost all studies used a study-specific questionnaire or instrument specifically designed to assess depressive problems (97.43%). All studies (100%) utilized appropriate statistical analysis to assess the prevalence of depressive problems. According to the Joanna Briggs institute quality assessment checklist, the included articles had a mean quality score of 7.71 ranging from five to nine. Twenty-five studies (64.10%) were high-quality studies (scored 7.71 and above) and the remaining were fair quality studies (scored between 5 and 7.71). (Table 2).

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The prevalence of depressive symptoms in homeless people (meta-analysis)

The pooled prevalence estimates of depressive symptoms in the homeless people was found to be 46.72% (95% CI; 37.77-55.90). The heterogeneity across the studies was significant for this analysis (I^2=97.57%; p <0.001) **(figure 2).**

The prevalence of major depressive disorders in homeless people (meta-analysis)

The pooled prevalence of major depressive disorders was 26.24% (95%CI 21.02-32.22). Significant heterogeneity has been found for this analysis (I^2=97.57; p <0.001) **(figure 3).**

The prevalence of persistent depressive disorders (dysthymia) disorder in homeless people

Our random-effects meta-analysis resulted in pooled prevalence estimates of dysthymia in the homeless people 8.25% (95% CI; 4.79-11.86). This analysis also resulted in a significant heterogeneity between the studies (I^2=89.80%; p <0.001) **(figure 4).**

Sensitivity analysis

Table 3 illustrates the results of the subgroup and sensitivity analysis. We conducted a sensitivity analysis by the age of the participants, the sample size, the instrument used to measure depressive problems, and the quality of the included studies in order to explore the possible source of heterogeneity in the analysis of the prevalence of depressive problems in the homeless people. All the sensitivity analyses were conducted by using a random effect model.

Our analysis based on the age of the participants demonstrated that the prevalence of depressive symptom was high among younger homeless people (<25 years of age), whereas the prevalence of major depressive disorder was high among older homeless people (>50 years of age) when compared with adults (25-50 years). The prevalence of depressive symptoms was 58.21% for younger participants and it was 44.76% and 47.44% for adult and older participants respectively. In these sensitivity analyses, the observed difference in the magnitude of depressive symptoms by the age of the participants was not statistically significant (P=0.261).

Similarly to depressive symptoms, we found no significant difference in the prevalence of depressive disorders by the age of the participants (P=0.839). The prevalence of depressive

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disorder was 22.98% for younger participants and it was 25.38% and 29.89% for adult and older participants respectively.

In our sensitivity analysis based on the quality of included studies, the pooled estimated prevalence of depressive symptoms was 40.45% for high-quality studies and it was 57.03 % for fair-quality studies, although the difference observed was not statistically significant (P=0.059**).** Likewise, the prevalence of depressive disorder was 21.75% as measured for high-quality studies and it was 32.45 % for fair-quality studies, although the difference observed was statistically significant (P=0.002**) (Table 2).**

We also found that the prevalence of depressive disorders was highest when measured by MINI (33.64%) followed by CIDI (27.42%), SCID (21.66%), and DSM (20.18%). However, the observed differences across the tools were not statistically significant (P>0.005).

Moreover, our sensitivity analysis based on the sample size revealed that the prevalence estimates of both depressive symptoms as well as depressive disorders were found to considerably higher in studies that involved a lower small sample size when compared with studies that involved a higher sample size.

Finally, one of the studies included in the final analysis was conducted among female participants only. When we remove this study from the final analysis, the prevalence estimates remained virtually the same 26.13% (95%CI 20.76-32.33).

Publication bias

In this review, the funnel plot was symmetric and regression tests associated with the funnel plot (Egger's test) provided no evidence of potential publication bias for the prevalence of depressive symptoms (B=5.11, SE=35.78, P=0.390) as well as depressive disorders (B=0.289, SE=3.26, P=0.930). **(Figure 5& 6).**

Discussion

To our knowledge, this is the first comprehensive systematic review and meta-analysis on the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people. We found 39 studies measuring the three types of depressive problems among homeless

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people: depressive symptoms, major depressive disorders, and dysthymia. Our analysis resulted in a considerably higher prevalence of depressive symptoms, (46.72%), major depressive disorders (26.24%), as well as dysthymia (8.25%) among homeless people. These prevalence rates are notably higher compared to that of the general population, suggesting depressive symptoms, major depressive disorders, and dysthymia are important and global public health issues among homeless people requiring urgent attention in terms of prevention and treatments.

The review showed that the existing literature on the prevalence of depressive symptoms, major depressive disorders, and dysthymia, among homeless people demonstrated a significant variation depending on the age of the participants and the reported quality of the studies. The vast majority of the included studies were performed in high-income countries (92.5%; $n = 37$) and only a few studies were performed in low and middle-income countries (7.5%; $n = 3$). , Almost all studies used standard instruments to measure the prevalence of depressive symptoms, major depressive disorders, and dysthymia among the homeless.

The results of our systematic review and meta-analysis suggest that nearly half of homeless people have experienced depressive symptoms. Thus, the prevalence of depressive symptoms among homeless people is 6.5 times higher than the estimates in the general population (7.2%) [18]. One of the possible explanations for this difference could be the higher prevalence rates of acute and chronic medical conditions, including HIV/AIDS, tuberculosis, and other medical problems among the homeless people that possibly increase the risks of psychiatric disorders including depressive symptoms as reported in many previous studies [61-66]. Another possible explanation is that homeless people are more likely to experience sexual assault and physical abuse that is frequently and consistently associated with a higher risk of depressive problems among the exposed individuals [67-70]. Finally, when compared with the general population the magnitude of substance use disorders, as well as other psychiatric problems, are considerably high among homeless people, which are associated with greater risks of depressive symptoms among the homeless [1, 2, 7, 52, 71, 72].

Our study showed that more than one-fourth of the homeless people had major depressive disorders, which was 2.30 times higher than the estimates from a previous systematic review conducted in Western countries among the homeless (11.4%) [73]. The reason for variation in the prevalence could be due to: (1) the previous study was conducted only in Western countries and

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the current study included studies from developed as well as developing countries. Thus, the potential reason for the higher in the prevalence of depressive disorders in the current study may be due to the inclusion of new studies including studies from low and middle-income countries. (2), the variation in characteristics of the participants across the countries in terms of experiencing the potential contributing factors that increase the risk of major depressive disorders.

Surprisingly, the prevalence of major depressive disorders in the current study was 5.25 times higher than the global prevalence of MDD in the general population reported by a recent meta-analysis (5%) [19]. Similarly, pooled prevalence estimates in the current study 1.5 to 3 times higher than. the pooled prevalence estimates of dysthymia in the general population (3-6%) [74]. As discussed earlier the possible factors for the higher prevalence in homeless people could be due to the presence of stressful life events, as well as comorbid medical, mental, and substance use disorders among the homeless people [1, 2, 7, 52, 61-72].

This review also showed that the prevalence of depressive symptoms was high among younger homeless people (<25 years of age), whereas the prevalence of major depressive disorder was high among older homeless people (>50 years of age) when compared with adults (25-50 years). The findings support the views that depressive disorders are common in older people than younger adults and they are more likely to be severe in presentation, display cognitive changes, and more complex presentations with high somatic symptoms [75]. The possible risk factors for these include age-related neurobiological changes, cognitive diathesis, and increased rates of stressful life events [76-78].

Strength and limitations

Estimating the prevalence of the different categories of depressive problems separately (depressive symptoms, dysthymia, and major depressive disorders) and assessing the prevalence across the lifespan are the main strengths of the present systematic review and meta-analysis.

Nevertheless, our study has some limitations: firstly, the vast majority of the included studies were performed in high-income countries and only three studies were conducted in low and middle-income countries, which significantly affects the global representativeness of the estimates. Secondly, we included only peer-reviewed published articles so that book reviews, gray literature, editorials, and letters were excluded from the study during the selection process. Thirdly, studies

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published in the English language are only included, which suggests that relevant studies published in another language might be missed. In fact, our evaluation of possible publication bias (small study effects due to possible missingness) revealed that the funnel plot was symmetric and Egger's regression tests provided no evidence of substantial publication bias for the prevalence of depressive symptoms, dysthymia, as well as major depressive disorders among the homeless people. Fourthly, the observed heterogeneity across the studies used to measure depressive problems was the other limitation of the current meta-analysis. To further detect the potential source of heterogeneity we conducted a sensitivity analysis by the age of the participants, the sample size, the instrument used to measure the outcome, and the quality of the included studies. Our analysis revealed that the quality of the studies and the sample size explained some of the observed heterogeneity ($P<0.05$). Nevertheless, the results should be interpreted with caution. Fifthly, we were unable to conduct sensitivity analysis stratified by the type of comorbid mental or substance use disorders, trauma-experiences, as well comorbid medical conditions including HIV/AIDS due to a limited number of studies in each subgroup.

The other limitation of the study is that only one reviewer (GA) conducted the selection of the studies. Epidemiological evidence indicates that the involvement of more than one reviewer in the entire process could increase the probability of including more studies in the final review [79]. However, this is not a major concern in our systematic review and meta-analysis since our assessment of publication bias showed no evidence of significant publication bias or small study effects for the prevalence of depressive symptoms (B=5.11, SE=35.78, P=0.390) as well as depressive disorders (B=0.289, SE=3.26, P=0.930), suggesting the effects of missed studies (if any) were not statistically significant.

The implication of the findings

Our findings suggest the urgent need for robust studies to explore the possible reasons for the higher prevalence of depressive symptoms, depressive disorders, and dysthymia among homeless people as compared to the reported magnitude in the general population. In addition, further studies to estimate the prevalence of depressive problems in low and middle-income countries are warranted. Finally, given the high prevalence rates of depressive problems and their potential negative consequences, early screening and interventions should be considered among those population groups. The intervention strategies might include the integration of homelessness,

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mental health, physical health, as well as alcohol and drug use, and other human services, working in a coordinated and joined-up way to deliver tailored and holistic interventions for those population groups.

Conclusion

This systematic review and meta-analysis revealed that the prevalence of depressive symptoms, dysthymia, and depressive disorders are notably high among homeless people, underlying the importance of early screening and preventions for depressive problems among the homeless people. Further strong studies are needed to examine the possible reasons for the higher prevalence of depressive problems among homeless people. Also, future robust and informative studies need to investigate better mechanisms of prevention, screening, and detection management of those problems among homeless people.

Abbreviations

AIDS: Acquired Immune Deficiency Syndrome

BDI: Beck's Depression Inventory

DASS: Depression, Anxiety, and Stress Scale

CIDI: Composite International Neuropsychiatric Interview

DSM IV-TR Diagnostic Statistical Manual of Mental disorders fourth, Edition, text revised

DSM-5-TR: Diagnostic Statistical Manual of Mental disorders fifth, Edition, text revised HIV: Human Immune Deficiency Virus

ICD: International Classification of Disease

MINI: Mini International Neuropsychiatric Interview

NOS: Newcastle-Ottawa Scale

PHQ: Patient Health Questionnaire

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

PSE: Present State Examination

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RBSD: Rost–Burnam Screener for Depression

UK: United Kingdom

USA: United States of America

WHO: World Health Organization

Declarations

Ethics approval and consent to participate

N/A

Consent for publication

N/A

Availability of data and material

All data generated or analyzed during this study are included in this article.

Competing interests

The authors declare that there are no competing interests.

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Authors' contributions

The author (GA) conceptualized the study, conducted analyses, write-up, and approval of the final manuscript. AB was performed the search, data extraction, quality assessment, write the discussion, draft, and approved the final manuscript. BD and BA were involved in reviewing the methodology and the results. They also read and approved the final manuscript. LT was participated in a critical review of the results as well as discussion and consensus and approved the final manuscript.

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587

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Table 1. Distribution of studies on depression in homeless people included in the qualitative and quantitative analysis based on year, study design, sample size, instrument, country, response rate, study population, and prevalence.

Author (year) (reference number)	Sample size	Tools used	Country	Study population	Outcome (magnitude of depressive problems)
Landefeld et al, 2017 [21]	348	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Old adults age greater than= 50	53.5%(n=185)
Bacciardi S et al, 2018 [30]	489	Mini-International Neuropsychiatric Interview (MINI)	USA	Adult with TBI	35.78%(n=175)
Rogers et al. (2017) [31]	124	Rost–Burnam Screener for Depression (RBSD)	USA	Adults≥50	52(n=64)
Lee KH et.al 2017 [13]	156	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	ADULTS	120 (77.5%)
Palar K et.al (2015) [32]	346	Beck Depression Interview (BDI)	USA	Adults with HIV	139 (35.8%)
Noel F et al. (2016) [33]	497	Mini-International Neuropsychiatric Interview (MINI)	Canada	Adults, MENTALLY ILL	40%(n=199)
Roze et.al. (2018) [34]	733	Composite International Neuropsychiatric Interview (CIDI)	France	Women Adults	28.8%
Sarajlija et.al. (2014) [35]	104	Beck Depression Inventory (BDI)	Serbia	Adults	15.4%(n=16) MDD 62.5% (n=65/104) BDI-depressive symptoms
Berg et al.2005 [14]	415	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adults(18-55) with latent TB	50%(n=209)
Topolovec-Vranic 2017 [20]	990	Mini International Neuropsychiatric Interview (MINI)	Canada	Adults with mental lines/TBI	42.1% (417/990
Coohey Y C et,=.al (2015) [16]	457	Diagnostic Statistical Manual of Mental disorders (DSM)	USA	Adults	26.25%(n=)

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Pluck, G.(2008) [36]	50	Zung Self-rating Depression Scale	UK	Adults	62.14%(n=31)
Fletcher JB et.al. (2017) [37]	131	Structured Clinical Interview for DSM-IV Axis I Disorders (SCID)	USA	Adults who were have sex with men(MSM)	39%(n=51)
La Gory et.al (1990) [38]	150	Centre for Epidemiologic Surveys for Depression (CES-D)	UK	Adult	75% (N=150)
Susser et.al. (1989) [39]	223	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adults(Men)	33%(n=74)
Munoz et.al (1998) [40]	262 Madrid	CIDI (Composite International Neuropsychiatric Interview)	Spain	Adult	MDD Madrid 21% (55/262) MDD los Angeles 21.2 (332/1563)
	1563 Los Angeles	Diagnostic Interview Schedule (DIS)			Dysthemia madrid 17.7 47/262) Los Angelees 14.8 (231)
Nyamathi et.al. (2012) [41]	267	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adult (18-46) with G/B	62.2%(n= 166)
Ghose T. et.al (2013) [42]	2838	International Classification of Disease	USA	Adults VETRIAN	9.5%(n=269)
Okamura T et.al (2014) [43]	423	Patient Health Questionnaire (PHQ2(two question Instrument))	JAPAN	adults	28.9%(N=119 /412)
Brown et.al (2013) [15]	472	Patient Health Questionnaire (PHQ-9)	USA	adults aged 50 years and older	39.8%(N=99)
Logan et.al 2013 [44]	208	Not mentioned	USA	young adults aged 18–25 years	64.4(N=134)
Rhoades et.al (2013) [45]	305	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adult men heterosexual(age > 18)	46.36%(n=14 1)
Strehlua et.al (2012) [46]	196	Mini International Neuropsychiatric Interview (MINI)	CANADA	Women with age 19-25	29%(n=56)
Prinsloo et.al (2012) [47]	38	Structured Clinical Interview for DSM-IV Axis I Disorders (SCID)	Iran	Adult with age ≥ 26	42.1%(n=16

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Whitbeck et.al. (2012) [48]	112	Composite International Neuropsychiatric Interview (CIDI)	Canada	ADULT	29.7%(n=33)
Hadland et.al (2011)[3]	559	Centre for Epidemiologic Surveys for Depression (CES-D)	CANADA	Young with age 14-26	43.4%(n=194)
Crawford et.al (2011)[17]	241	Composite International Neuropsychiatric Interview (CIDI)	USA	Young women with age b/n 16 and 19	32.50%(n=73)
Nyamathi et.al (212) [49]	156	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Young Adults	52.60%(n=82)
Larney et.al (2009) [50]	105	DASS (Depression, Anxiety, and Stress Scale)	Australia	Adult	44.76%(n=47)
Greifenhage et.al (1997) [51]	36	Diagnostic Statistical Manual of Mental disorders (DSM)	Germany	Adult women	MDD 50%(n=16) Dysthymia 6 %(n=2)
Koegel et.al 1998 [52]	445	Diagnostic Statistical Manual of Mental disorders (DSM)	USA	Adult	MDD 18.3%(n=60) Dysthymia 9.3% (n=41)
Fichter et.al (2001) [53]	265	Structured Clinical Interview for DSM-IV Axis I Disorders (SCID)	Germany	Adult men	21.1%(n=59)
Kovess et.al (1999) [54]	838	Composite International Neuropsychiatric Interview (CIDI)	France	Adult	33.7%(n=241)
Rohde et.al. (2001) [55]	523	Structured Clinical Interview for DSM-IV Axis I Disorders (SCID)	USA	Adolescent age less than 21	Depression 17.6%(n=92) MDD 12.2% (N=61) Dysthymia 6.5% (n=34)
Spinelli et.al (2017) [56]	350	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adult with age 50 and above	38.3%(n=134)
Van Straaten et al 2014 [57]	385	Depression, Anxiety, and Stress Scale (DASS)	Netherlands	All homeless	41.55(160)
Bassuk el et.al 1986 [58]	80	Diagnostic Statistical Manual of Mental disorders (DSM)	USA	Adult	12.5% (n=10)
Herman H et.al , 1989 [59]	382	Diagnostic Statistical Manual of Mental disorders (DSM)	Australia	Adult	20% (n=78)

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Ayano et.al 2017 [1]	456	Diagnostic Statistical Manual of Mental disorders (DSM)	Ethiopia	adult	9.6% (n=44)
Krausz RM et.al, 2013	489	Mini International Neuropsychiatric Interview (MINI)	CANADA	-	MDD 45.429%(n=222) Dysthymia 4.7 (n=23)

Key: UK: United Kingdom; USA: United States of America

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Table 2: Qualities of studies included in the systematic review and meta-analysis

Study name	Response									
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Total
Landefeld et al, 2017 [21]	Y	N	Y	Y	Y	Y	Y	Y	U	7
Bacciardi S et al, 2018 [30]	Y	N	Y	Y	Y	Y	Y	Y	U	7
Rogers et al. (2017) [31]	Y	N	N	Y	Y	N	Y	Y	U	6
Lee KH et.al 2017 [13]	Y	N	N	Y	Y	N	Y	Y	U	6
Palar K et.al (2015) [32]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Noel F et al. (2016) [33]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Roze et.al. (2018) [34]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Sarajlija et.al. (2014) [35]	Y	N	Y	Y	Y	Y	Y	Y	Y	8
Berg et al.2005 [14]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Topolovec-Vranic 2017 [20]	N	Y	Y	Y	Y	Y	Y	Y	U	7
Coohey Y C et,=.al (2015) [16]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Pluck, G.(2008) [36]	N	N	N	Y	Y	Y	Y	Y	U	5
Fletcher JB et.al. (2017) [37]	N	N	N	Y	Y	Y	Y	Y	U	5
La Gory et.al (1990) [38]	Y	Y	N	Y	Y	Y	Y	Y	U	7
Susser et.al. (1989) [39]	Y	Y	N	Y	Y	Y	Y	Y	U	7
Munoz et.al (1998) [40]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Nyamathi et.al. (2012) [41]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Ghose T. et.al (2013) [42]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Okamura T et.al (2014) [43]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Brown et.al (2013) [15]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Logan et.al 2013 [44]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Rhoades et.al (2013) [45]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Strehlua et.al (2012) [46]	N	Y	Y	Y	Y	Y	Y	Y	U	8
Prinsloo et.al (2012) [47]	N	Y	N	Y	Y	Y	Y	Y	Y	7
Whitbeck et.al. (2012) [48]	N	Y	N	Y	Y	Y	Y	Y	U	6
Hadland et.al (2011)[3]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Crawford et.al (2011)[17]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Nyamathi et.al (212) [49]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Larney et.al (2009) [50]	N	Y	N	Y	Y	Y	Y	Y	Y	7
Greifenhage et.al (1997) [51]	N	Y	N	Y	Y	Y	Y	Y	Y	7
Koegel et.al 1998 [52]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Fichter et.al (2001) [53]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Kovess et.al (1999) [54]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Rohde et.al. (2001) [55]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9

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Spinelli et.al (2017) [56]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Van Straaten et al 2014 [57]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Bassuk el et.al 1986 [58]	N	Y	N	Y	Y	Y	Y	Y	U	6
Herman H et.al , 1989 [59]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Ayano et.al 2017 [1]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9

Keys:

A. Q1-Q9 represents questions used to assess the quality of included studies, which are listed below.

Q1. Was the sample frame appropriate to address the target populations?

Q2. Were the study participants sampled in appropriate way?

Q3. Was the sample size adequate?

Q4. Were the study subjects and setting described in details?

Q5. Was the data analysis conducted with sufficient coverage of the identified sample?

Q6. Was a valid method used in the identification of conditions?

Q7. Was the condition measured in a standard, reliable way for all participants?

Q8. Was there an appropriate statistical analysis?

Q9. Was the response rate adequate, and if not, was the low response rate managed appropriately?

Y, yes; N, no; U, unclear; NA, not applicable

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598 **Table 3:** Subgroup and sensitivity analysis of prevalence of depressive problems among homeless
599 participants by age and quality of the studies

Subgroup	**Number of studies**	**Estimates**		Heterogeneity across the studies		Heterogeneity between groups (*P* value
		Prevalence (%)	**95% Confidence interval**	**I^2 (%)**	**P-value**	
Age (depressive symptom)						
Adult	13	44.76	32.95-57.20	98.04	<0001	0.261
Older	3	47.44	37.22-57.88	88.14	<0001	
Younger	2	58.21	46.13-69.38	88.60	<0001	
Age (depressive disorders)						
Adult	16	25.38	19.44-32.40	97.72	<0001	0.839
Older	2	29.89	14-84-51.05	95.28	<0001	
Younger	3	22.98	11.96-39.58	95.79	<0001	
Quality of the study (depressive symptoms)						
High	11	40.45	29.68-52.23	93.30	<0.0001	0.059
Fair	7	57.03	44.39-68.81	98.17	<0.0001	
Quality of the study (depressive disorders)						
High	14	21.75	16.67-27.86	81.24	<0.0001	0.002
Fair	7	35.45	29.31-42.11	97.43	<0.0001	
Sample size (depressive symptoms)						
300 and above	14	37.30	27.05-48.82	98.09	<0.0001	0.008
300 and above	14	59.18	47.74-69.69	93.41	<0.0001	
Sample size (depressive disorders)						

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300 and above	14	23.91	17.99-31.04	98.23	<0.0001	0.397
below 300	8	27.81	22.12-34.32	79.69	<0.0001	

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Figure legends

Figure 1: **PRISMA flowchart of review search**

This figure illustrates the process of searching for relevant studies from the three reputable databases including identification, selection, eligibility and inclusion of the studies depending on the predefined criteria**.**

Figure 2: Forest plot of the prevalence of depressive symtoms among homeless people: a meta-analysis

The figure shows the results of the meta-analysis of the studies on depressive symtoms among homeless people using random-effect model.

Figure 3: Forest plot of the prevalence of major depressive disorders among homeless people: a meta-analysis

The figure shows the results of the meta-analysis of the studies on major depressive disorders among homeless people using random-effect model.

Figure 4: Forest plot of the prevalence of dysthemia among homeless people: a meta-analysis

The figure shows the results of the meta-analysis of the studies on dysthemia among homeless people using random-effect model.

Figure 5: Publication bias for depressive symtoms among homeless people

Figure 6: Publication bias for major depressive disorders among homeless people

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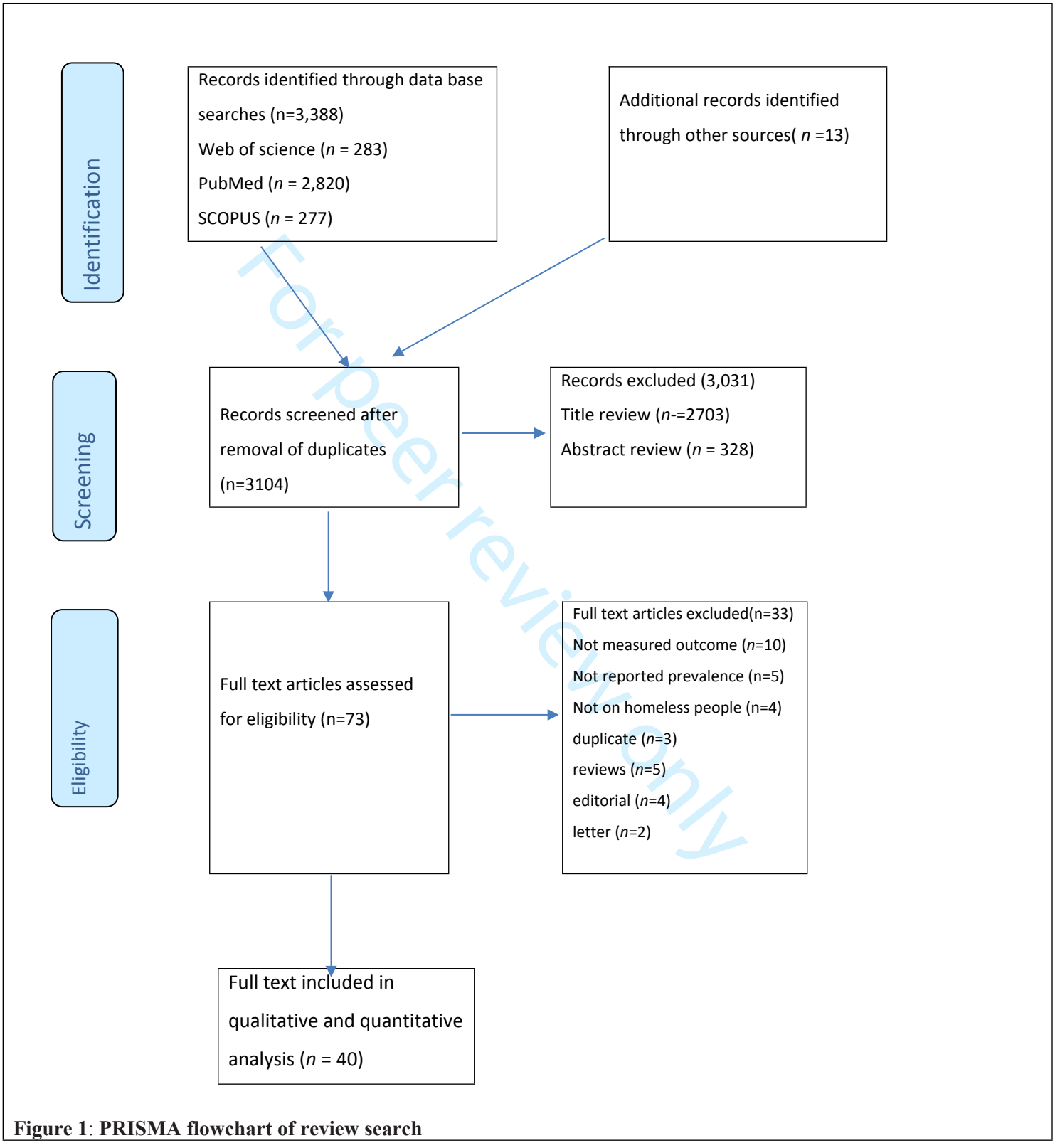
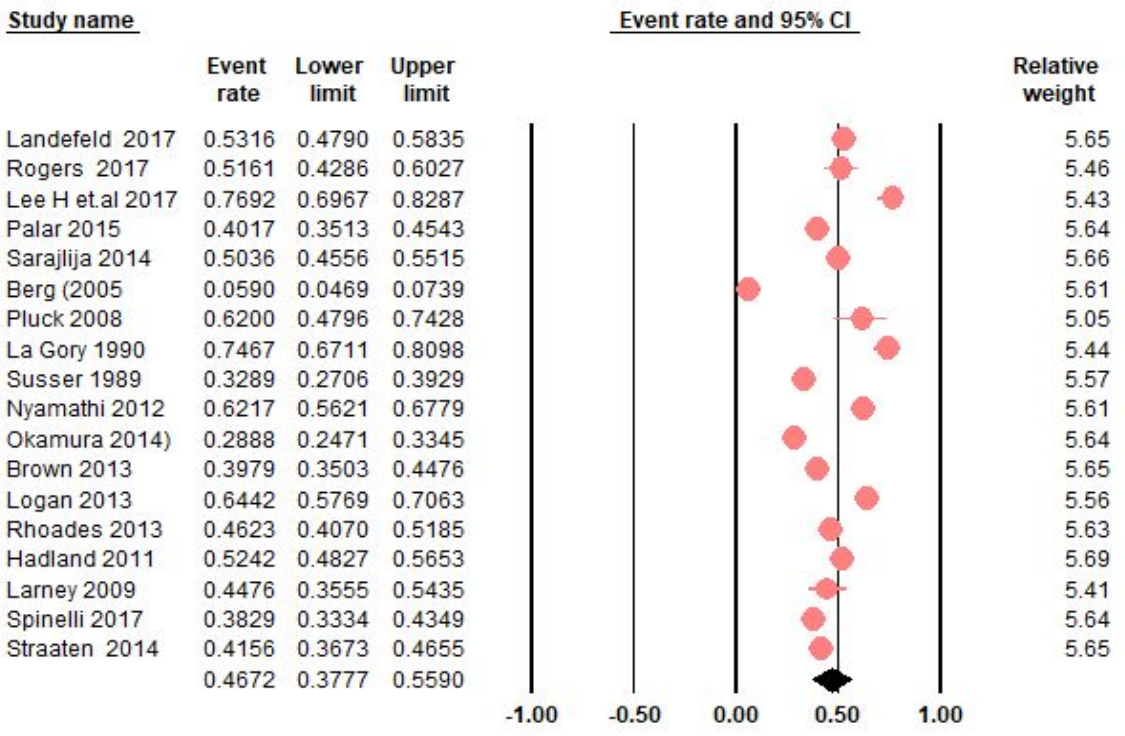


Figure 1: **PRISMA flowchart of review search**

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Figure 2: The prevalence of depressive symptoms among homeless people: a meta-analysis

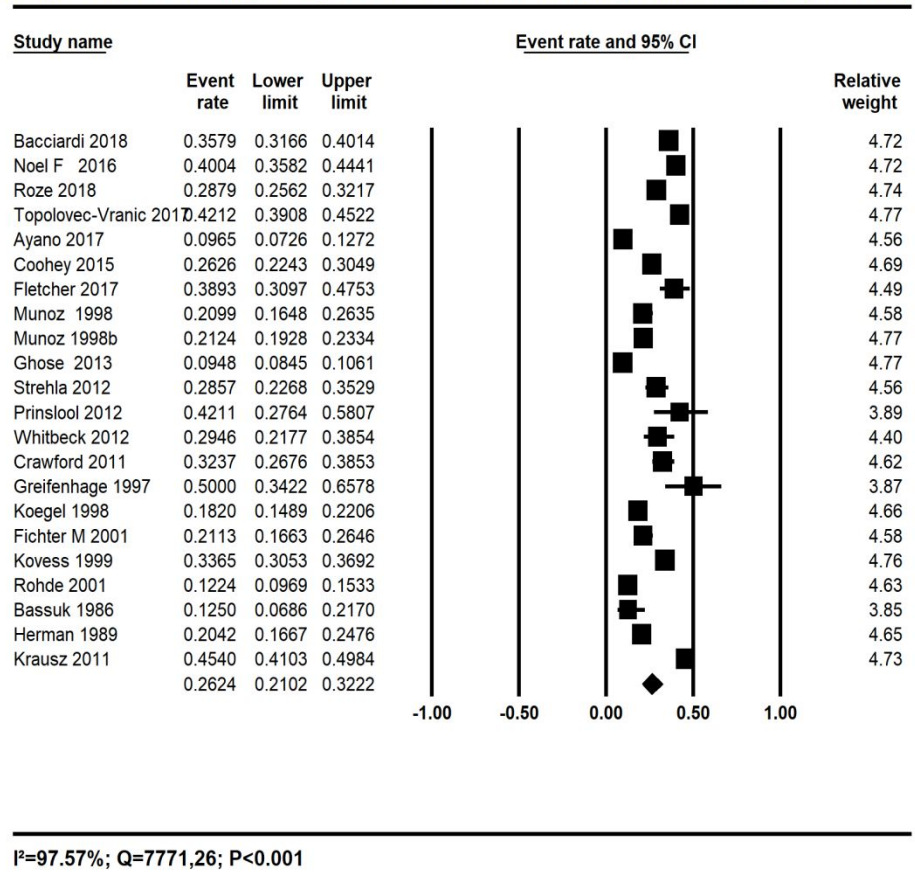


Keys: I^2=97.57%, Q=698.54; df=17; P<0.0001; Based on random effect analysis

w only

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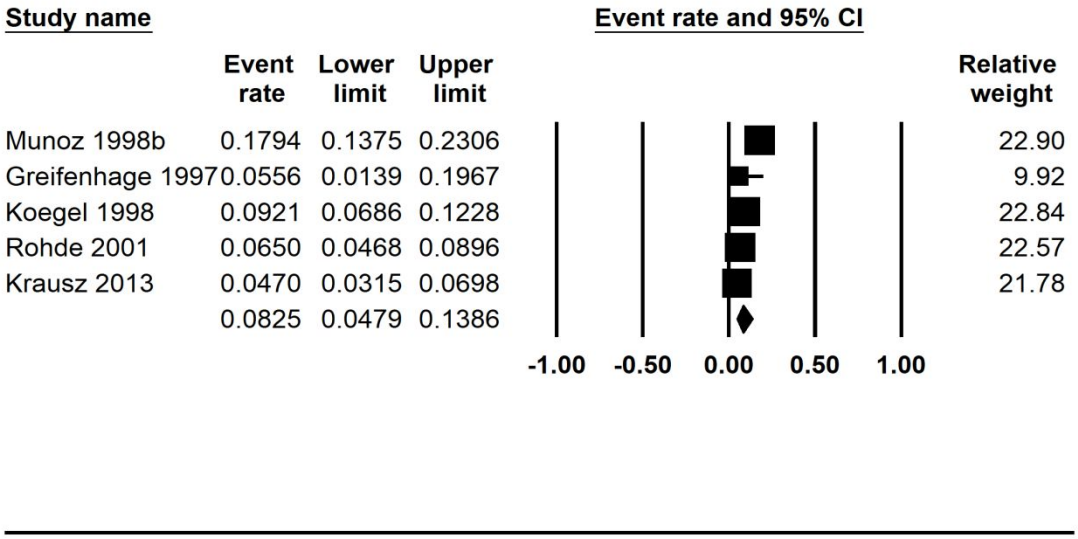
Figure 3: The prevalence of major depressive disorders among homeless people: a meta-analysis



I^2=97.57%; Q=7771,26; P<0.001

w only

Figure 4: The prevalence of dysthymia among homeless people: a meta-analysis



I^2=89.80%; Q=39.20; P<0.001

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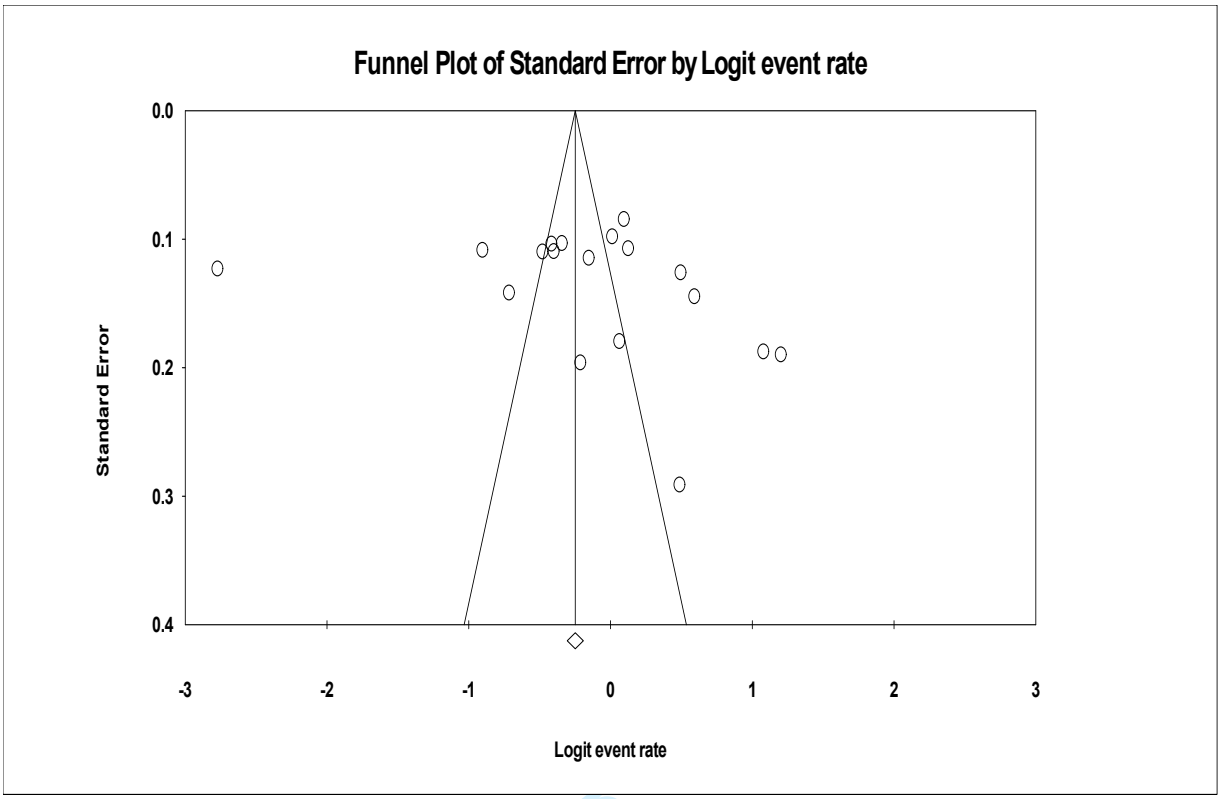


Figure 5: The risk of publication bias for the prevalence of depressive symptoms among homeless people

review only

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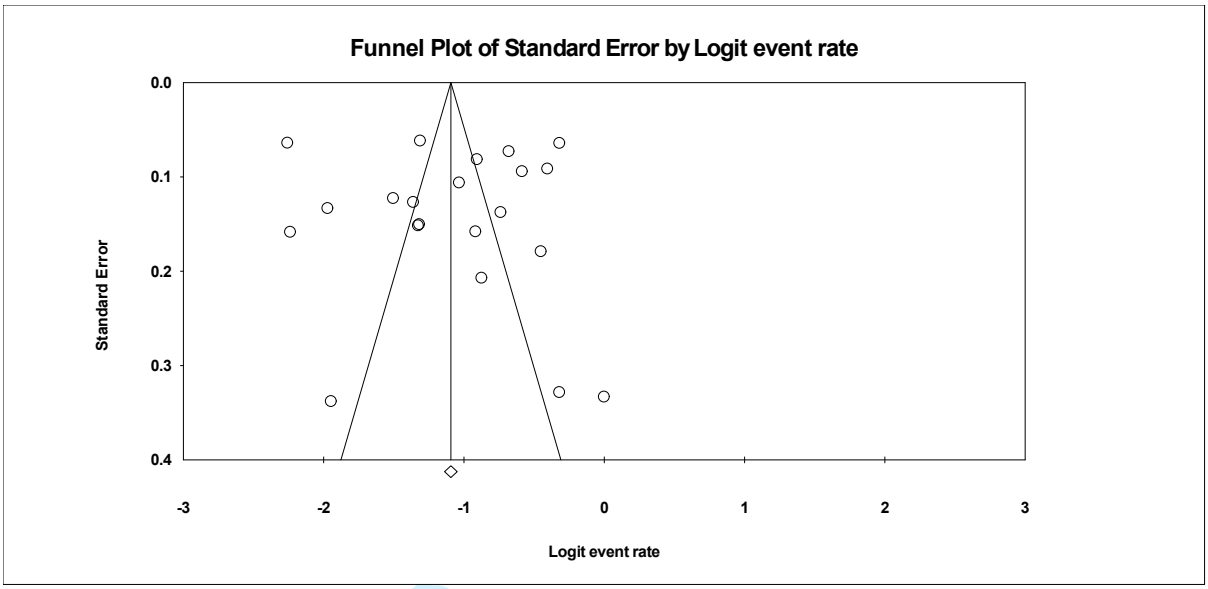


Figure 6: The risk of publication bias for the prevalence of major depressive disorders among homeless

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Supplementary material 1: Full list of databases searching terms used

Databases	**Searching terms**	Results
PubMed	((("depressive disorder"[MeSH Terms] OR ("depressive"[All Fields] AND "disorder"[All Fields]) OR "depressive disorder"[All Fields] OR "depression"[All Fields] OR "depression"[MeSH Terms]) OR ("depression"[MeSH Terms] OR "depression"[All Fields] OR ("depressive"[All Fields] AND "symptoms"[All Fields]) OR "depressive symptoms"[All Fields]) OR ("depressive disorder"[MeSH Terms] OR ("depressive"[All Fields] AND "disorder"[All Fields]) OR "depressive disorder"[All Fields]) OR ("mental disorders"[MeSH Terms] OR ("mental"[All Fields] AND "disorders"[All Fields]) OR "mental disorders"[All Fields] OR ("mental"[All Fields] AND "disorder"[All Fields]) OR "mental disorder"[All Fields]) OR ("dysthymic disorder"[MeSH Terms] OR ("dysthymic"[All Fields] AND "disorder"[All Fields]) OR "dysthymic disorder"[All Fields] OR "dysthymia"[All Fields]) OR ("mental disorders"[MeSH Terms] OR ("mental"[All Fields] AND "disorders"[All Fields]) OR "mental disorders"[All Fields] OR ("psychiatric"[All Fields] AND "disorder"[All Fields]) OR "psychiatric disorder"[All Fields]) OR ("mental disorders"[MeSH Terms] OR ("mental"[All Fields] AND "disorders"[All Fields]) OR "mental disorders"[All Fields] OR ("mental"[All Fields] AND "illness"[All Fields]) OR "mental illness"[All Fields]) OR (("mental health"[MeSH Terms] OR ("mental"[All Fields] AND "health"[All Fields]) OR "mental health"[All Fields]) AND problems[All Fields])) AND (("homeless persons"[MeSH Terms] OR ("homeless"[All Fields] AND "persons"[All Fields]) OR "homeless persons"[All Fields] OR "homeless"[All Fields]) OR ("homeless persons"[MeSH Terms] OR ("homeless"[All Fields] AND "persons"[All Fields]) OR "homeless persons"[All Fields] OR "homelessness"[All Fields]))) AND (("epidemiology"[Subheading] OR "epidemiology"[All Fields] OR "prevalence"[All Fields] OR "prevalence"[MeSH Terms]) OR magnitude [All Fields]).	2820

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Scopus	Depression* OR 'depressive symptom' OR 'depressive disorder' OR dysthymia* OR homeless* OR homelessness* OR prevalence* OR magnitude*	277
Web of Science	**TOPIC:** (depression OR depressive symptom OR depressive disorder OR dysthymia) *AND* **TOPIC:** (homeless OR homelessness) *AND* **TOPIC:** (prevalence OR magnitude) **Timespan:** All years. **Indexes:** SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC.	283

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4-5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7-8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7-8
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7-9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9-11
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	13

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

A systematic review and meta-analysis of the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people

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Primary Subject Heading:	Mental health
Secondary Subject Heading:	Complementary medicine, Public health
Keywords:	Adult psychiatry < PSYCHIATRY, PSYCHIATRY, Depression & mood disorders < PSYCHIATRY, PUBLIC HEALTH

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A systematic review and meta-analysis of the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people

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Abstract

Objectives

To assess the global prevalence estimates of depressive symptoms, dysthymia, and major depressive disorders among homeless people.

Design

Systematic review and meta-analysis

Data sources

Databases including PubMed, Scopus, and Web of Science were systematically searched up to February 2020 to identify relevant studies that have reported data on the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people.

Eligibility criteria

Original epidemiologic studies written in English that addressed the prevalence of depressive problems among homeless people

Data extraction and synthesis

A random-effect meta-analysis was performed to pool the prevalence estimated from individual studies. Subgroup and sensitivity analyses were employed to compare the prevalence across the groups as well as to identify the source of heterogeneities. The Joanna Briggs Institute's quality assessment checklist was used to measure the study quality. Cochran's Q and the I^2 test were used to assess heterogeneity between the studies.

Results

Forty publications, including 17,215 participants, were included in the final analysis. This meta-analysis demonstrated considerably higher prevalence rates of depressive symptoms 46.72% (95% CI; 37.77-55.90), dysthymia 8.25% (95% CI; 4.79-11.86), as well as major depressive disorders 26.24% (95%CI 21.02-32.22) among homeless people. Our subgroup analysis showed that the prevalence of depressive symptoms was high among younger homeless people (<25 years of age), whereas the prevalence of major depressive disorder was high among older homeless people (>50 years of age) when compared with adults (25-50 years).

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Conclusion

This review showed that nearly half, one-fourth, and one-tenth of homeless people are suffering from depressive symptoms, dysthymia, and major depressive disorders, respectively, which are notably higher than the reported prevalence rates in the general population. The findings suggest the need for appropriate mental health prevention and treatment strategies for this population group.

Keywords: Homeless, depressive symptoms, dysthymia, major depressive disorders, systematic review, meta-analysis

Strength and limitations of this study

- Estimating the prevalence of the different categories of depressive problems separately (depressive symptoms, dysthymia, and major depressive disorders) and assessing the prevalence across the lifespan are the main strengths of the present systematic review and meta-analysis.
- The vast majority of the included studies were performed in high-income countries and only three studies were conducted in low and middle-income countries, which significantly affects the global representativeness of the estimates.
- We included only peer-reviewed published articles.
- Only studies published in the English language were included in this meta-analysis, therefore, relevant studies published in another language might have been missed.

Background

Global epidemiological evidence shows that a substantial proportion of homeless people living in both high income and low-income countries are suffering from mental, neurologic, and substance use disorders [1-3]. The reported prevalence estimates of mental, neurologic, and substance use disorders among homeless people ranged from 25% to 92 % [4-7]. Research evidence indicates that the presence of mental and substance disorders among homeless people is frequently and consistently associated with a higher risk of negative outcomes including disability and mortality from alcohol and drug use, suicide, and general medical conditions [8-12].

Regarding depressive problems, in particular, scientific evidence from numerous studies has shown that depressive problems are the commonest mental health problems among homeless

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people [13-17]. The reported prevalence estimates of depressive problems among the homeless people ranged from 9.5 to 76.9% across the studies [13-17]. These prevalence rates are higher than the estimates in the general population (5-7.2%) [18, 19]. Moreover, the vast majority of homeless people with depressive problems have comorbid physical disease, mental, and substance use disorders [1, 3, 20]. Epidemiological evidence suggests that comorbid conditions in patients with depressive disorders are associated with more severe disability, suffering, suicide, as well as higher mortality rates from different causes when compared with non-comorbid depressive disorders [8-11, 21-25].

However, to the best of our knowledge, there is no previous systematic review and meta-analysis that measured the global prevalence estimates of depressive symptoms, dysthymia, and major depressive disorders among homeless people. This study fills this gap in the literature by analyzing the global prevalence estimates of those depressive problems among homeless people. The finding from such analysis will provide robust data on depressive problems among homeless individuals, which in turn helps policy-makers and program managers in developing policy solutions and designing appropriate and effective intervention strategies for those population groups.

Methods

Data sources and search strategies

This systematic review and meta-analysis was performed in accordance with the guidelines of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [26]. We used a predesigned protocol (unpublished) for searching, extracting data, inclusion or exclusion of studies, assessing quality, and data analysis. A systematic search of relevant studies was carried out in three reputable databases ─PubMed, Scopus, and Web of Science using keywords and combinations tailored to each database. The details about the searching technique are in supplementary material 1. In addition, the reference lists of included studies were hand-searched to identify additional articles.

Eligibility criteria and study selection

In this systematic review and meta-analysis, studies were included if they: (1) were conducted among homeless people; (2) assessed the prevalence of depressive symptoms, dysthymia, or major

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depressive disorders or reported data to estimate the prevalence and (3) written in English language. Studies were excluded if they satisfy the following criteria: (1) they were commentaries, reviews, case reports, and animal studies; (2) they were letters, editorials, conference papers, books, and notes.

Selection of studies for the inclusion in the systematic review and meta-analysis

In this systematic review and meta-analysis, the corresponding author (GA) identified articles and consequently evaluated them by their titles and abstract based on the eligibility criteria. Full-text articles were selected for further assessment by the author. This author further appraised the full text of each study and subsequently retained those full-text articles to be included in the final analysis.

Definition of key terms and concepts

Depressive disorders and dysthymia

This systematic review and meta-analysis combined studies from a wide range of settings and countries across the globe that reported the lifetime prevalence estimate of major depressive disorders as well as dysthymia according to standard diagnostic criteria's, which were assessed by using standard diagnostic instruments such as the Diagnostic Statistical Manual of Mental disorders (DSM), Schedule for Clinical Assessment of Neuropsychiatry (SCAN), International Classification of Disease (ICD), and Composite International Neuropsychiatric Interview (CIDI) and Mini-International Neuropsychiatric Interview (MINI).

Depressive symptoms

The prevalence of depressive symptoms in this review indicates the current prevalence as measured by the screening instruments such as the Centre for Epidemiologic Surveys for Depression (CES-D), Depression, Anxiety, and Stress Scale (DASS), Patient Health Questionnaire (PHQ-9), and Zung Self-rating Depression Scale.

Homelessness

There is no internationally consistent definition of homelessness. Some scholars (countries) define homelessness narrowly as the absence of access to safe shelter while others define broadly comprising those people living in marginal housing (marginal accommodations), people who are sleeping in public places or designed shelter, and rooflessness [27, 28]. In the present study, we

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have included studies that defined homelessness in various ways. Thus, homelessness represents the comprehensive definition mentioned above (which is beyond rooflessness).

Methods for data extraction and quality assessment

Two independent authors (AB and GA) performed data extraction from the included studies. We extracted the following data from each study: first author(s) name, country of the study, sample size, year of publication, tools used, and a number of positive cases, and the reported prevalence of depressive symptoms, dysthymia, and major depressive disorders. The Joanna Briggs institute quality assessment tool was used to assess the quality of studies included in this meta-analysis [29]. According to this scale, the scoring of each study was conducted based on the frequency scales that are answered as yes, no, not clear (when some information is missing in the study), and not applicable (the question does not apply to that study depending on the design. The total quality score for each study was calculated based on the total number of positive scores.

Data synthesis and analysis

All statistical analysis was conducted by using comprehensive meta-analysis software version 3. The prevalence and 95% confidence interval were calculated using a random-effect model [30]. The I^2 statistics was used to measure between-study heterogeneity [30]. The values of I^2 statistics such as 25, 50, and 75% represented low, medium, and high heterogeneity respectively [31]**.** The ages of the participants and the quality of the studies was used to determine the possible source of heterogeneity between the studies. Publication bias was assessed by using the funnel plot and Egger's regression tests. For all analyses, the P-value for statistical significance was set at 0.05.

Patient and public involvement

Patient and the public were not involved in this study.

Results

Identification of relevant studies

After database searches performed strictly according to a predesigned protocol, 3388 studies were identified through an electronic database search. Further 13 studies have been added using manual searches making the total studies 3401. Of these, we excluded 2703 articles during the assessment of duplicate and titles, as they did not satisfy the selection criteria. Subsequently, 328 articles were

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excluded after the assessment of the abstract as they did not meet the inclusion criteria. Hence, a full-text of 73 articles were retained for further evaluation and 40 of these were found to be eligible for meta-analysis. (Figure 1)

Characteristics of included studies

Overviews of the key characteristics of the studies included in this systematic review and meta-analysis are shown in Table 1. A total of 40 studies [1, 3, 13-17, 20, 21, 32-62], with 17,215 participants, were included in the analysis. Majority of these studies (n = 21) were conducted in the USA [13-17, 21, 32-34, 39, 41, 43, 44, 46, 47, 51, 54, 57, 58], and six were conducted in Canada [3, 20, 35, 48, 50, 62], two conducted in Australia [52, 61], two conducted in Germany [53, 55], two conducted in the UK [38, 40], two conducted in France[36, 56], one conducted in Spain [42], one conducted in Japan[45], one conducted in Iran [49], one conducted in Serbia [37], one conducted in Netherlands [59], and one conducted in Ethiopia [1]. The eligible articles were published between 1989 and 2017. The sample size for the included studies ranged from 36 participants in Germany to 2838 participants in the USA.

Regarding the specific outcomes, 21 studies assessed depressive symptoms, 20 studies assessed major depressive disorders, and 5 studies assessed dysthymia (chronic minor depressive disorders).

Depressive symptoms among homeless people were predominantly measured using the Center for Epidemiologic Studies Depression Scale-Revised (CES-D). The CES-D was used in 11 studies, Beck's Depression Inventory (BDI) in 2 studies, Patient Health Questionnaire (PHQ) in 2 studies, Depression, Anxiety, and Stress Scale (DASS), Rost–Burnam Screener for Depression (RBSD)in one study, and one study used a self-reported history of depressive symptoms.

Regarding major depressive disorders, the Majority (n =11) of the studies used the Diagnostic Statistical Manual of Mental disorders (DSM) to measure major depressive disorders. While five studies used Composite International Neuropsychiatric Interview (CIDI), five studies used Mini-International Neuropsychiatric Interview (MINI), and International Classification of Disease (ICD) applied in one study. All studies used DSM to measure dysthymia.

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Quality of included studies

Table 2 demonstrates the risk of bias and the quality of the included articles. Twenty-eight studies utilized an adequate sample size (71.79%) to measure the prevalence of depressive problems. About one-third of the studies scored positive on the item regarding response rate (35.89%) and the remaining studies scored negative (64.11%). To measure depressive problems, the vast majority of the studies utilized a standard instrument or valid diagnostic criteria (97.43%), and almost all studies used a study-specific questionnaire or instrument specifically designed to assess depressive problems (97.43%). All studies (100%) utilized appropriate statistical analysis to assess the prevalence of depressive problems. According to the Joanna Briggs institute quality assessment checklist, the included articles had a mean quality score of 7.71 ranging from five to nine. Twenty-five studies (64.10%) were high-quality studies (scored 7.71 and above) and the remaining were fair quality studies (scored between 5 and 7.71). (Table 2).

The prevalence of depressive symptoms in homeless people (meta-analysis)

The pooled prevalence estimates of depressive symptoms in the homeless people was found to be 46.72% (95% CI; 37.77-55.90). The heterogeneity across the studies was significant for this analysis (I^2=97.57%; p <0.001) **(figure 2).**

The prevalence of major depressive disorders in homeless people (meta-analysis)

The pooled prevalence of major depressive disorders was 26.24% (95%CI 21.02-32.22). Significant heterogeneity has been found for this analysis (I^2=97.57; p <0.001) **(figure 3).**

The prevalence of persistent depressive disorders (dysthymia) disorder in homeless people

Our random-effects meta-analysis resulted in pooled prevalence estimates of dysthymia in the homeless people 8.25% (95% CI; 4.79-11.86). This analysis also resulted in a significant heterogeneity between the studies (I^2=89.80%; p <0.001) **(figure 4).**

Sensitivity analysis

Table 3 illustrates the results of the subgroup and sensitivity analysis. We conducted a sensitivity analysis by the age of the participants, the sample size, the instrument used to measure depressive problems, and the quality of the included studies in order to explore the possible source of

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heterogeneity in the analysis of the prevalence of depressive problems in the homeless people. All the sensitivity analyses were conducted by using a random effect model.

Our analysis based on the age of the participants demonstrated that the prevalence of depressive symptom was high among younger homeless people (<25 years of age), whereas the prevalence of major depressive disorder was high among older homeless people (>50 years of age) when compared with adults (25-50 years). The prevalence of depressive symptoms was 58.21% for younger participants and it was 44.76% and 47.44% for adult and older participants respectively. In these sensitivity analyses, the observed difference in the magnitude of depressive symptoms by the age of the participants was not statistically significant (P=0.261).

Similarly to depressive symptoms, we found no significant difference in the prevalence of depressive disorders by the age of the participants (P=0.839). The prevalence of depressive disorder was 22.98% for younger participants and it was 25.38% and 29.89% for adult and older participants respectively.

In our sensitivity analysis based on the quality of included studies, the pooled estimated prevalence of depressive symptoms was 40.45% for high-quality studies and it was 57.03 % for fair-quality studies, although the difference observed was not statistically significant (P=0.059**).** Likewise, the prevalence of depressive disorder was 21.75% as measured for high-quality studies and it was 32.45 % for fair-quality studies, although the difference observed was statistically significant (P=0.002**)** **(Table 2).**

We also found that the prevalence of depressive disorders was highest when measured by MINI (33.64%) followed by CIDI (27.42%), SCID (21.66%), and DSM (20.18%). However, the observed differences across the tools were not statistically significant (P>0.05).

Moreover, our sensitivity analysis based on the sample size revealed that the prevalence estimates of both depressive symptoms as well as depressive disorders were found to considerably higher in studies that involved a lower small sample size when compared with studies that involved a higher sample size.

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Finally, one of the studies included in the final analysis was conducted among female participants only. When we remove this study from the final analysis, the prevalence estimates remained virtually the same 26.13% (95%CI 20.76-32.33).

Publication bias

In this review, the funnel plot was symmetric and regression tests associated with the funnel plot (Egger's test) provided no evidence of potential publication bias for the prevalence of depressive symptoms (B=5.11, SE=35.78, P=0.390) as well as depressive disorders (B=0.289, SE=3.26, P=0.930). **(Figure 5& 6).**

Discussion

To our knowledge, this is the first comprehensive systematic review and meta-analysis on the prevalence of depressive symptoms, dysthymia, and major depressive disorders among homeless people. We found 39 studies measuring the three types of depressive problems among homeless people: depressive symptoms, major depressive disorders, and dysthymia. Our analysis resulted in a considerably higher prevalence of depressive symptoms, (46.72%), major depressive disorders (26.24%), as well as dysthymia (8.25%) among homeless people. These prevalence rates are notably higher compared to that of the general population, suggesting depressive symptoms, major depressive disorders, and dysthymia are important and global public health issues among homeless people requiring urgent attention in terms of prevention and treatments.

The review showed that the existing literature on the prevalence of depressive symptoms, major depressive disorders, and dysthymia, among homeless people demonstrated a significant variation depending on the age of the participants and the reported quality of the studies. The vast majority of the included studies were performed in high-income countries (92.5%; n = 37) and only a few studies were performed in low and middle-income countries (7.5%; n = 3). , Almost all studies used standard instruments to measure the prevalence of depressive symptoms, major depressive disorders, and dysthymia among the homeless.

The results of our systematic review and meta-analysis suggest that nearly half of homeless people have experienced depressive symptoms. Thus, the prevalence of depressive symptoms among homeless people is 6.5 times higher than the estimates in the general population (7.2%) [18]. One of the possible explanations for this difference could be the higher prevalence rates of acute and

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chronic medical conditions, including HIV/AIDS, tuberculosis, and other medical problems among the homeless people that possibly increase the risks of psychiatric disorders including depressive symptoms as reported in many previous studies [63-68]. Another possible explanation is that homeless people are more likely to experience sexual assault and physical abuse that is frequently and consistently associated with a higher risk of depressive problems among the exposed individuals [69-72]. Finally, when compared with the general population the magnitude of substance use disorders, as well as other psychiatric problems, are considerably high among homeless people, which are associated with greater risks of depressive symptoms among the homeless [1, 2, 7, 54, 73, 74].

Our study showed that more than one-fourth of the homeless people had major depressive disorders, which was 2.30 times higher than the estimates from a previous systematic review conducted in Western countries among the homeless (11.4%) [75]. The reason for variation in the prevalence could be due to: (1) the previous study was conducted only in Western countries and the current study included studies from developed as well as developing countries. Thus, the potential reason for the higher prevalence of depressive disorders in the current study may be due to the inclusion of new studies including studies from low and middle-income countries. (2), the variation in characteristics of the participants across the countries in terms of experiencing the potential contributing factors that increase the risk of major depressive disorders.

Surprisingly, the prevalence of major depressive disorders in the current study was 5.25 times higher than the global prevalence of MDD in the general population reported by a recent meta-analysis (5%) [19]. Similarly, pooled prevalence estimates in the current study 1.5 to 3 times higher than. the pooled prevalence estimates of dysthymia in the general population (3-6%) [76]. As discussed earlier the possible factors for the higher prevalence in homeless people could be due to the presence of stressful life events, as well as comorbid medical, mental, and substance use disorders among the homeless people [1, 2, 7, 54, 63-74].

This review also showed that the prevalence of depressive symptoms was high among younger homeless people (<25 years of age), whereas the prevalence of major depressive disorder was high among older homeless people (>50 years of age) when compared with adults (25-50 years). The findings support the views that depressive disorders are common in older people than younger adults and they are more likely to be severe in presentation, display cognitive changes, and more complex presentations with high somatic symptoms [77]. The possible risk factors for these

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include age-related neurobiological changes, cognitive diathesis, and increased rates of stressful life events [78-80].

Strength and limitations

Estimating the prevalence of the different categories of depressive problems separately (depressive symptoms, dysthymia, and major depressive disorders) and assessing the prevalence across the lifespan are the main strengths of the present systematic review and meta-analysis.

Nevertheless, our study has some limitations: firstly, the vast majority of the included studies were performed in high-income countries and only three studies were conducted in low and middle-income countries, which significantly affects the global representativeness of the estimates. Secondly, we included only peer-reviewed published articles so book reviews, grey literature, editorials, and letters were excluded from the study during the selection process. Thirdly, studies published in the English language are only included, which suggests that relevant studies published in another language might be missed. In fact, our evaluation of possible publication bias (small study effects due to possible missingness) revealed that the funnel plot was symmetric and Egger's regression tests provided no evidence of substantial publication bias for the prevalence of depressive symptoms, dysthymia, as well as major depressive disorders among the homeless people. Fourthly, the observed heterogeneity across the studies used to measure depressive problems was the other limitation of the current meta-analysis. To further detect the potential source of heterogeneity we conducted a sensitivity analysis by the age of the participants, the sample size, the instrument used to measure the outcome, and the quality of the included studies. Our analysis revealed that the quality of the studies and the sample size explained some of the observed heterogeneity ($P<0.05$). Nevertheless, the results should be interpreted with caution. Fifthly, we were unable to conduct sensitivity analysis stratified by the type of comorbid mental or substance use disorders, trauma-experiences, as well comorbid medical conditions including HIV/AIDS due to a limited number of studies in each subgroup.

The other limitation of the study is that only one reviewer (GA) conducted the selection of the studies. However, this is not a major concern in our systematic review and meta-analysis since our assessment of publication bias showed no evidence of significant publication bias or small study effects for the prevalence of depressive symptoms ($B=5.11$, $SE=35.78$, $P=0.390$) as well as

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depressive disorders (B=0.289, SE=3.26, P=0.930), suggesting the effects of missed studies (if any) were not statistically significant. In addition, we have involved two independent investigators for data extraction and quality assessment. The existing guidelines in systematic review and meta-analysis including Cochrane library and PRISMA suggest quality assessment and data extraction must be performed by two independent investigators [26, 81]. However, involving the second author in entire processes including selection and screening is not mandatory (optional) [26, 81]. In fact, a recent systematic review indicated that the involvement of more than one reviewer in the entire process could increase the probability of including more studies in the final review [82]. Another systematic review on the subject revealed no significant differences in the rates of effectiveness (avoiding the number of studies with inappropriate exclusions) in studies that involved a second author in the entire process and studies that did not [83].

The implication of the findings

Our findings suggest the urgent need for robust studies to explore the possible reasons for the higher prevalence of depressive symptoms, depressive disorders, and dysthymia among homeless people as compared to the reported magnitude in the general population. In addition, further studies to estimate the prevalence of depressive problems in low and middle-income countries are warranted. Finally, given the high prevalence rates of depressive problems and their potential negative consequences, early screening and interventions should be considered among those population groups. The intervention strategies might include the integration of homelessness, mental health, physical health, as well as alcohol and drug use, and other human services, working in a coordinated and joined-up way to deliver tailored and holistic interventions for those population groups.

Conclusion

This systematic review and meta-analysis revealed that the prevalence of depressive symptoms, dysthymia, and depressive disorders are notably high among homeless people, underlying the importance of early screening and prevention for depressive problems among homeless people. Further strong studies are needed to examine the possible reasons for the higher prevalence of depressive problems among homeless people. Also, future robust and informative studies need to

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investigate better mechanisms of prevention, screening, and detection management of those problems among homeless people.

Abbreviations

AIDS: Acquired Immune Deficiency Syndrome

BDI: Beck's Depression Inventory

DASS: Depression, Anxiety, and Stress Scale

CIDI: Composite International Neuropsychiatric Interview

DSM IV-TR Diagnostic Statistical Manual of Mental disorders fourth, Edition, text revised

DSM-5-TR: Diagnostic Statistical Manual of Mental disorders fifth, Edition, text revised HIV: Human Immune Deficiency Virus

ICD: International Classification of Disease

MINI: Mini International Neuropsychiatric Interview

NOS: Newcastle-Ottawa Scale

PHQ: Patient Health Questionnaire

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

PSE: Present State Examination

RBSD: Rost–Burnam Screener for Depression

UK: United Kingdom

USA: United States of America

WHO: World Health Organization

Declarations

Ethics approval and consent to participate

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N/A

Consent for publication

N/A

Availability of data and material

All data generated or analyzed during this study are included in this article.

Competing interests

The authors declare that there are no competing interests.

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Authors' contributions

The author (GA) conceptualized the study, conducted analyses, write-up, and approval the final manuscript. AB was performed the search, data extraction, quality assessment, write the discussion, draft, and approved the final manuscript. BD and BA were involved in reviewing the methodology and the results. They also read and approved the final manuscript. LT participated in a critical review of the results as well as the discussion. All authors have approved the final manuscript.

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610

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Table 1. Distribution of studies on depression in homeless people included in the qualitative and quantitative analysis based on year, study design, sample size, instrument, country, response rate, study population, and prevalence.

Author (year) (reference number)	Sample size	Tools used	Country	Study population	Outcome (magnitude of depressive problems)
Landefeld et al, 2017 [21]	348	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Old adults age greater than= 50	Depressive symptoms 53.5%(n=185)
Bacciardi S et al, 2018 [32]	489	Mini-International Neuropsychiatric Interview (MINI)	USA	Adult with TBI	Depressive symptoms 35.78%(n=175)
Rogers et al. (2017) [33]	124	Rost–Burnam Screener for Depression (RBSD)	USA	Adults≥50	Depressive symptoms 52(n=64)
Lee KH et.al 2017 [13]	156	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	ADULTS	Depressive symptoms 120 (77.5%)
Palar K et.al (2015) [34]	346	Beck Depression Interview (BDI)	USA	Adults with HIV	Depressive symptoms 139 (35.8%)
Noel F et al. (2016) [35]	497	Mini-International Neuropsychiatric Interview (MINI)	Canada	Adults with Mental health problems	Depressive symptoms 40%(n=199)
Roze et.al. (2018) [36]	733	Composite International Neuropsychiatric Interview (CIDI)	France	Women (Adults)	MDD 28.8%
Sarajlija et.al. (2014) [37]	104	Beck Depression Inventory (BDI)	Serbia	Adults	MDD 15.4%(n=16)
					Depressive symptoms 62.5% (n=65/104)
Berg et al.2005 [14]	415	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adults(18-55) with latent TB	Depressive symptoms 50%(n=209)

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Topolovec-Vranic J et.al. (2017) [20]	990	Mini International Neuropsychiatric Interview (MINI)	Canada	Adults with mental helath problems /TBI	MDD 42.1% (417/990
Coohey Y C et,=.al (2015) [16]	457	Diagnostic Statistical Manual of Mental disorders (DSM)	USA	Adults	26.25%(n=)
Pluck, G.(2008) [38]	50	Zung Self-rating Depression Scale	UK	Adults	Depressive symptoms 62.14%(n=31)
Fletcher JB et.al. (2017) [39]	131	Structured Clinical Interview for DSM-IV Axis I Disorders (SCID)	USA	Adults who were have sex with men(MSM)	MDD 39%(n=51)
La Gory et.al (1990) [40]	150	Centre for Epidemiologic Surveys for Depression (CES-D)	UK	Adult	Depressive symptoms 75% (N=150)
Susser et.al. (1989) [41]	223	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adults(Men)	Depressive symptoms 33%(n=74)
Munoz et.al (1998) [42]	262 Madrid 1563 Los Angeles	CIDI (Composite International Neuropsychiatric Interview) Diagnostic Interview Schedule (DIS)	Spain	Adult	MDD Madrid 21% (55/262) MDD Los Angeles 21.2 (332/1563) Dysthemia madrid 17.7 47/262) Los Angelees 14.8 (231)
Nyamathi et.al. (2012) [43]	267	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adult (18-46) with G/B	Depressive symptoms 62.2%(n= 166)
Ghose T. et.al (2013) [44]	2838	International Classification of Disease (ICD)	USA	Adults (Veterans)	MDD 9.5%(n=269)
Okamura T et.al (2014) [45]	423	Patient Health Questionnaire (PHQ2(two question Instrument))	Japan	adults	Depressive symptoms 28.9%(N=119/ 412)
Brown et.al (2013) [15]	472	Patient Health Questionnaire (PHQ-9)	USA	adults aged 50 years and older	Depressive symptoms 39.8%(N=99)

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Logan et.al 2013 [46]	208	Not mentioned	USA	young adults aged 18–25 years	Depressive symptoms 64.4(N=134)
Rhoades et.al (2013) [47]	305	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adult men heterosexual(age > 18)	Depressive symptoms 46.36%(n=141)
Strehlua et.al (2012) [48]	196	Mini International Neuropsychiatric Interview (MINI)	Canada	Women with age 19-25	MDD 29%(n=56)
Prinsloo et.al (2012) [49]	38	Structured Clinical Interview for DSM-IV Axis I Disorders (SCID)	Iran	Adult with age > 26	MDD 42.1%(n=16
Whitbeck et.al. (2012) [50]	112	Composite International Neuropsychiatric Interview (CIDI)	Canada	ADULT	MDD 29.7%(n=33)
Hadland et.al (2011)[3]	559	Centre for Epidemiologic Surveys for Depression (CES-D)	Canada	Young with age 14-26	Depressive symptoms 43.4%(n=194)
Crawford et.al (2011)[17]	241	Composite International Neuropsychiatric Interview (CIDI)	USA	Young women with age b/n 16 and 19	MDD 32.50%(n=73)
Nyamathi et.al (212) [51]	156	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Young Adults	Depressive symptoms 52.60%(n=82)
Larney et.al (2009) [52]	105	DASS (Depression, Anxiety, and Stress Scale)	Australia	Adult	Depressive symptoms 44.76%(n=47)
Greifenhage et.al (1997) [53]	36	Diagnostic Statistical Manual of Mental disorders (DSM)	Germany	Adult women	MDD 50%(n=16) Dysthymia 6 %(n=2)
Koegel et.al 1998 [54]	445	Diagnostic Statistical Manual of Mental disorders (DSM)	USA	Adult	MDD 18.3%(n=60) Dysthymia 9.3% (n=41)
Fichter et.al (2001) [55]	265	Structured Clinical Interview for DSM-IV Axis I Disorders (SCID)	Germany	Adult men	MDD 21.1%(n=59)
Kovess et.al (1999) [56]	838	Composite International Neuropsychiatric Interview (CIDI)	France	Adult	MDD 33.7%(n=241)
Rohde et.al. (2001) [57]	523	Structured Clinical Interview for DSM-IV Axis I Disorders (SCID)	USA	Adolescent age less than 21	Depressive symptoms 17.6%(n=92) MDD 12.2% (N=61)

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					Dysthymia 6.5% (n=34)
Spinelli et.al (2017) [58]	350	Centre for Epidemiologic Surveys for Depression (CES-D)	USA	Adult with age 50 and above	Depressive symptoms 38.3%(n=134)
Van Straaten et al 2014 [59]	385	Depression, Anxiety, and Stress Scale (DASS)	Netherlands	All homeless	Depressive symptoms 41.55(160)
Bassuk el et.al 1986 [60]	80	Diagnostic Statistical Manual of Mental disorders (DSM)	USA	Adult	MDD 12.5% (n=10)
Herman H et.al , 1989 [61]	382	Diagnostic Statistical Manual of Mental disorders (DSM)	Australia	Adult	MDD 20% (n=78)
Ayano et.al 2017 [1]	456	Diagnostic Statistical Manual of Mental disorders (DSM)	Ethiopia	adult	MDD 9.6% (n=44)
Krausz RM et.al, 2013	489	Mini International Neuropsychiatric Interview (MINI)	Canada	-	MDD 45.429%(n=222) Dysthymia 4.7 (n=23)

Key: UK: United Kingdom; USA: United States of America, MDD: Major Depressive Disorders, ICD:

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618 **Table 2:** The quality of studies included in the systematic review and meta-analysis

Study name	Response									
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Total
Landefeld et al, 2017 [21]	Y	N	Y	Y	Y	Y	Y	Y	U	7
Bacciardi S et al, 2018 [32]	Y	N	Y	Y	Y	Y	Y	Y	U	7
Rogers et al. (2017) [33]	Y	N	N	Y	Y	N	Y	Y	U	6
Lee KH et.al 2017 [13]	Y	N	N	Y	Y	N	Y	Y	U	6
Palar K et.al (2015) [34]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Noel F et al. (2016) [35]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Roze et.al. (2018) [36]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Sarajlija et.al. (2014) [37]	Y	N	Y	Y	Y	Y	Y	Y	Y	8
Berg et al.2005 [14]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Topolovec-Vranic J et.al. (2017) [20]	N	Y	Y	Y	Y	Y	Y	Y	U	7
Coohey Y C et,=.al (2015) [16]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Pluck, G.(2008) [38]	N	N	N	Y	Y	Y	Y	Y	U	5
Fletcher JB et.al. (2017) [39]	N	N	N	Y	Y	Y	Y	Y	U	5
La Gory et.al (1990) [40]	Y	Y	N	Y	Y	Y	Y	Y	U	7
Susser et.al. (1989) [41]	Y	Y	N	Y	Y	Y	Y	Y	U	7
Munoz et.al (1998) [42]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Nyamathi et.al. (2012) [43]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Ghose T. et.al (2013) [44]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Okamura T et.al (2014) [45]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Brown et.al (2013) [15]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Logan et.al 2013 [46]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Rhoades et.al (2013) [47]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Strehlua et.al (2012) [48]	N	Y	Y	Y	Y	Y	Y	Y	U	8
Prinsloo et.al (2012) [49]	N	Y	N	Y	Y	Y	Y	Y	Y	7
Whitbeck et.al. (2012) [50]	N	Y	N	Y	Y	Y	Y	Y	U	6
Hadland et.al (2011)[3]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Crawford et.al (2011)[17]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Nyamathi et.al (212) [51]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Larney et.al (2009) [52]	N	Y	N	Y	Y	Y	Y	Y	Y	7
Greifenhage et.al (1997) [53]	N	Y	N	Y	Y	Y	Y	Y	Y	7
Koegel et.al 1998 [54]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Fichter et.al (2001) [55]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Kovess et.al (1999) [56]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Rohde et.al. (2001) [57]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9

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Spinelli et.al (2017) [58]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Van Straaten et al 2014 [59]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9
Bassuk el et.al 1986 [60]	N	Y	N	Y	Y	Y	Y	Y	U	6
Herman H et.al , 1989 [61]	Y	Y	Y	Y	Y	Y	Y	Y	U	8
Ayano et.al 2017 [1]	Y	Y	Y	Y	Y	Y	Y	Y	Y	9

Keys:

A. Q1-Q9 represents questions used to assess the quality of included studies, which are listed below.

Q1. Was the sample frame appropriate to address the target populations?

Q2. Were the study participants sampled in appropriate way?

Q3. Was the sample size adequate?

Q4. Were the study subjects and setting described in details?

Q5. Was the data analysis conducted with sufficient coverage of the identified sample?

Q6. Was a valid method used in the identification of conditions?

Q7. Was the condition measured in a standard, reliable way for all participants?

Q8. Was there an appropriate statistical analysis?

Q9. Was the response rate adequate, and if not, was the low response rate managed appropriately?

Y, yes; N, no; U, unclear; NA, not applicable

619

620

621

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Table 3: Subgroup and sensitivity analysis of prevalence of depressive problems among homeless participants by age and quality of the studies

Subgroup	Number of studies	Estimates		Heterogeneity across the studies		Heterogeneity between groups (*P* value
		Prevalence (%)	95% Confidence interval	I^2 (%)	P-value	
Age (depressive symptom)						
Adult	13	44.76	32.95-57.20	98.04	<0001	0.261
Older	3	47.44	37.22-57.88	88.14	<0001	
Younger	2	58.21	46.13-69.38	88.60	<0001	
Age (depressive disorders)						
Adult	16	25.38	19.44-32.40	97.72	<0001	0.839
Older	2	29.89	14-84-51.05	95.28	<0001	
Younger	3	22.98	11.96-39.58	95.79	<0001	
Quality of the study (depressive symptoms)						
High	11	40.45	29.68-52.23	93.30	<0.0001	0.059
Fair	7	57.03	44.39-68.81	98.17	<0.0001	
Quality of the study (depressive disorders)						
High	14	21.75	16.67-27.86	81.24	<0.0001	0.002
Fair	7	35.45	29.31-42.11	97.43	<0.0001	
Sample size (depressive symptoms)						
300 and above	14	37.30	27.05-48.82	98.09	<0.0001	0.008
300 and above	14	59.18	47.74-69.69	93.41	<0.0001	
Sample size (depressive disorders)						

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300 and above	14	23.91	17.99-31.04	98.23	<0.0001	0.397
below 300	8	27.81	22.12-34.32	79.69	<0.0001	

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Figure legends

Figure 1: **PRISMA flowchart of review search**

This figure illustrates the process of searching for relevant studies from the three reputable databases including identification, selection, eligibility and inclusion of the studies depending on the predefined criteria**.**

Figure 2: The prevalence of depressive sysmpoms among homeless people: a meta-analysis

The figure shows the results of the meta-analysis of the studies on depressive symtoms among homeless people using random-effect model.

Figure 3: The prevalence of major depressive disorders among homeless people: a meta-analysis

The figure shows the results of the meta-analysis of the studies on major depressive disorders among homeless people using random-effect model.

Figure 4: The prevalence of dyfthymia among homeless people: a meta-analysis

The figure shows the results of the meta-analysis of the studies on dysthemia among homeless people using random-effect model.

Figure 5: The risk of publication bias for the prevalence of depressive symptoms among homeless people

Figure 6: The risk of publication bias for the prevalence of major depressive disorders among homeless

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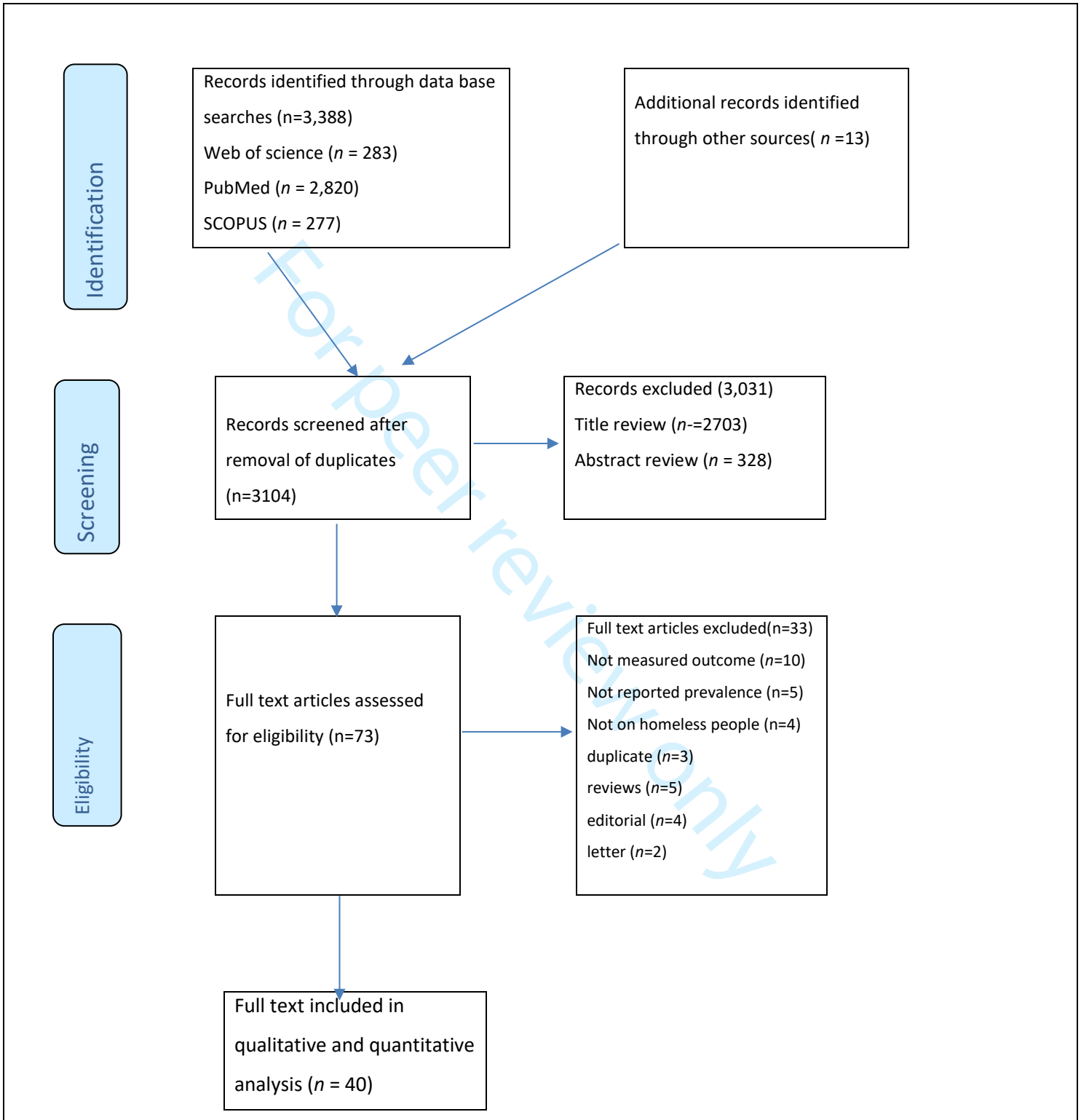
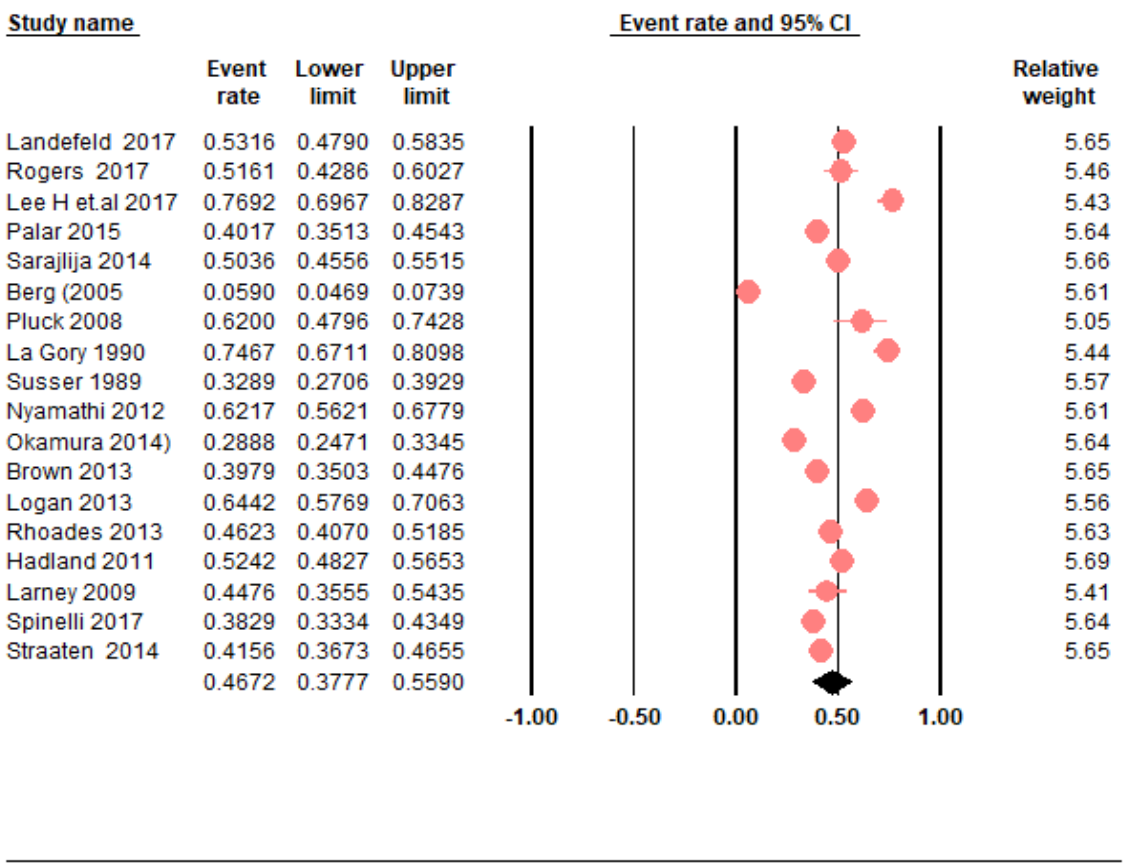


Figure 1: **PRISMA flowchart of review search**

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Figure 2: The prevalence of depressive symptoms among homeless people: a meta-analysis

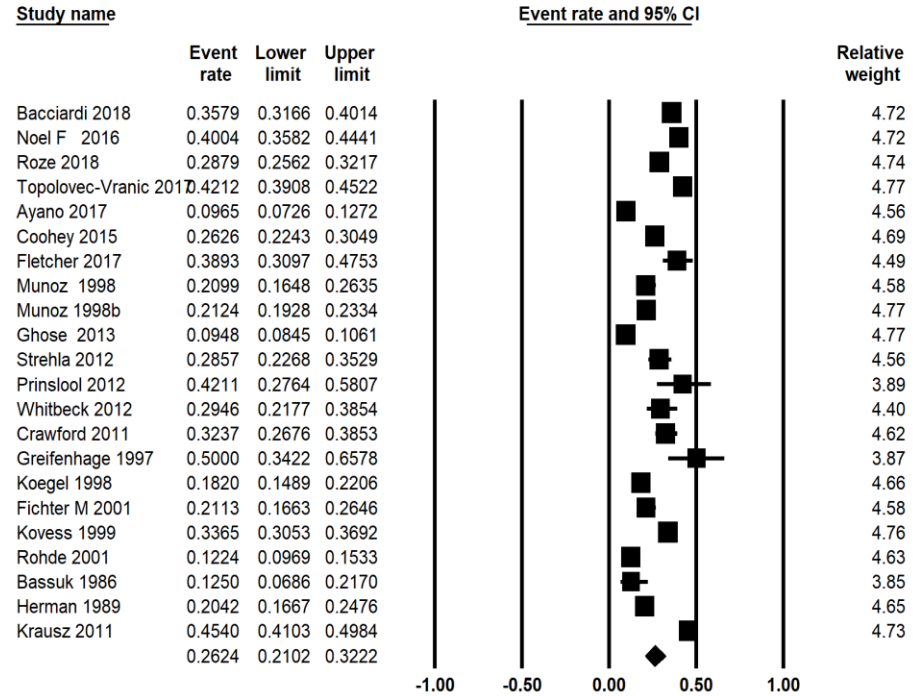


Keys: I^2=97.57%, Q=698.54; df=17; P<0.0001; Based on random effect analysis

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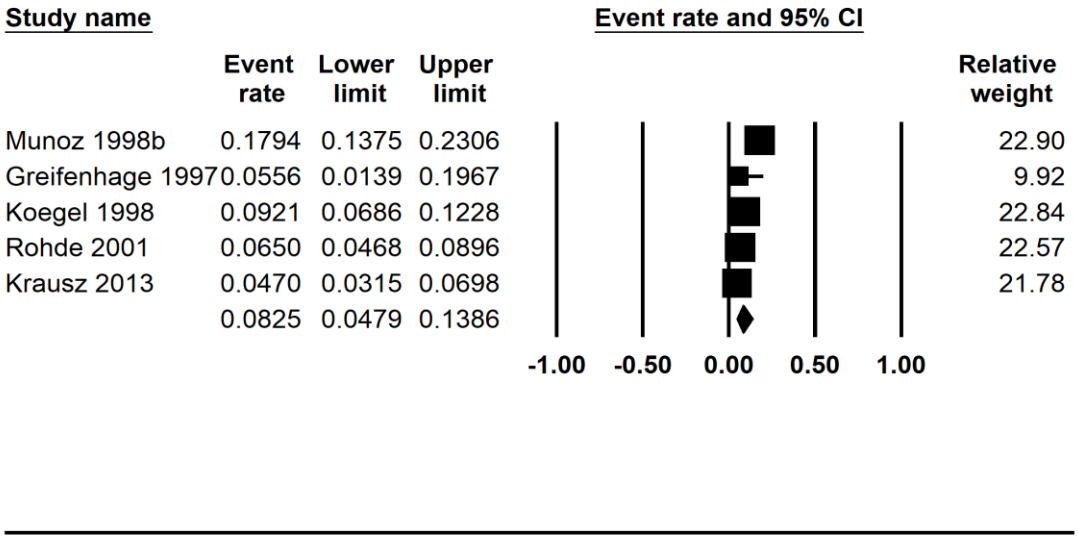
Figure 3: The prevalence of major depressive disorders among homeless people: a meta-analysis



I^2=97.57%; Q=7771,26; P<0.001

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Figure 4: The prevalence of dysthymia among homeless people: a meta-analysis



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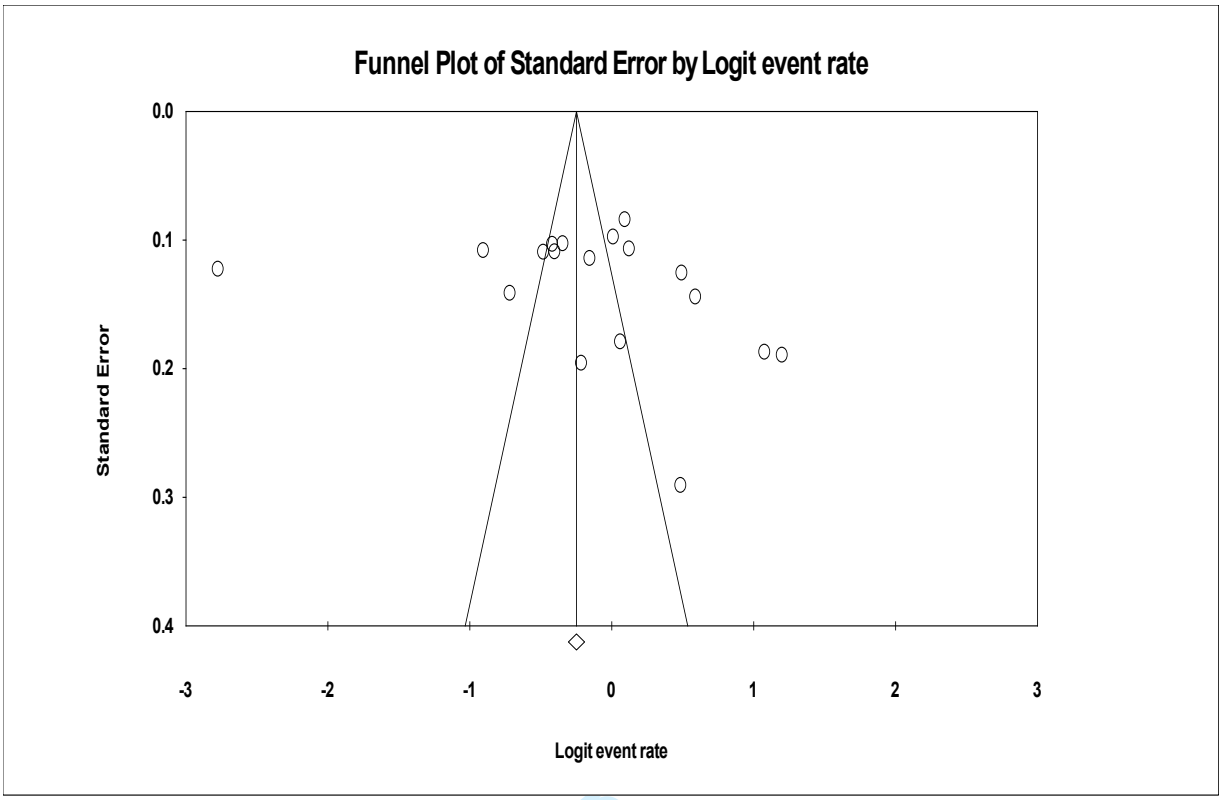


Figure 5: The risk of publication bias for the prevalence of depressive symptoms among homeless people

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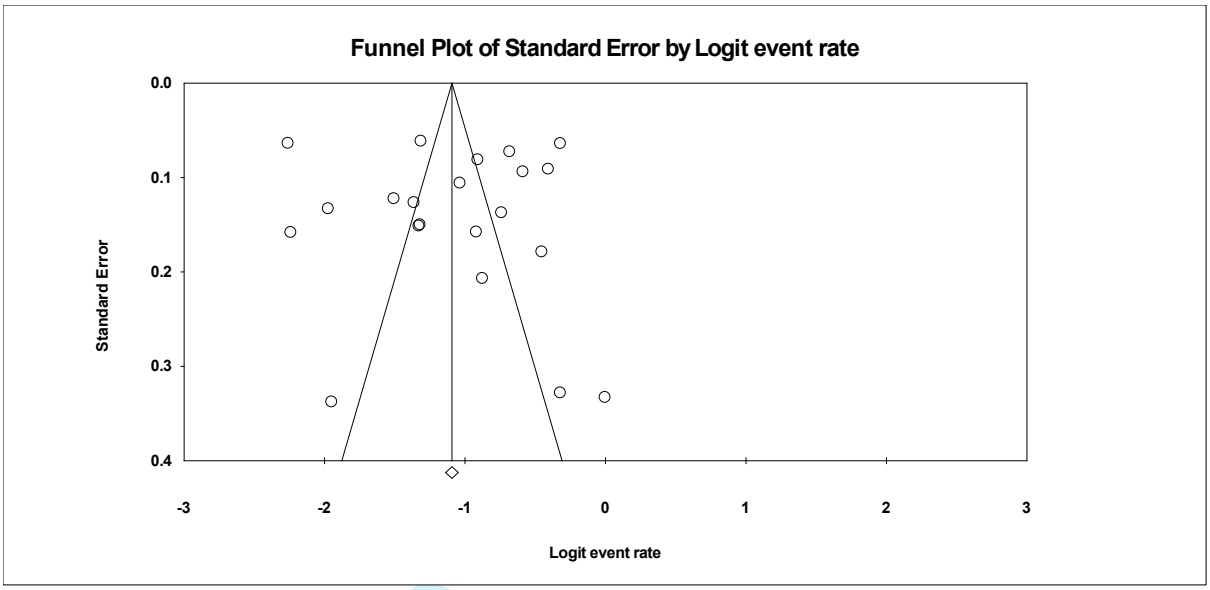


Figure 6: The risk of publication bias for the prevalence of major depressive disorders among homeless

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Supplementary material 1: Full list of databases searching terms used

Databases	**Searching terms**	Results
PubMed	((("depressive disorder"[MeSH Terms] OR ("depressive"[All Fields] AND "disorder"[All Fields]) OR "depressive disorder"[All Fields] OR "depression"[All Fields] OR "depression"[MeSH Terms]) OR ("depression"[MeSH Terms] OR "depression"[All Fields] OR ("depressive"[All Fields] AND "symptoms"[All Fields]) OR "depressive symptoms"[All Fields]) OR ("depressive disorder"[MeSH Terms] OR ("depressive"[All Fields] AND "disorder"[All Fields]) OR "depressive disorder"[All Fields]) OR ("mental disorders"[MeSH Terms] OR ("mental"[All Fields] AND "disorders"[All Fields]) OR "mental disorders"[All Fields] OR ("mental"[All Fields] AND "disorder"[All Fields]) OR "mental disorder"[All Fields]) OR ("dysthymic disorder"[MeSH Terms] OR ("dysthymic"[All Fields] AND "disorder"[All Fields]) OR "dysthymic disorder"[All Fields] OR "dysthymia"[All Fields]) OR ("mental disorders"[MeSH Terms] OR ("mental"[All Fields] AND "disorders"[All Fields]) OR "mental disorders"[All Fields] OR ("psychiatric"[All Fields] AND "disorder"[All Fields]) OR "psychiatric disorder"[All Fields]) OR ("mental disorders"[MeSH Terms] OR ("mental"[All Fields] AND "disorders"[All Fields]) OR "mental disorders"[All Fields] OR ("mental"[All Fields] AND "illness"[All Fields]) OR "mental illness"[All Fields]) OR (("mental health"[MeSH Terms] OR ("mental"[All Fields] AND "health"[All Fields]) OR "mental health"[All Fields]) AND problems[All Fields])) AND (("homeless persons"[MeSH Terms] OR ("homeless"[All Fields] AND "persons"[All Fields]) OR "homeless persons"[All Fields] OR "homeless"[All Fields]) OR ("homeless persons"[MeSH Terms] OR ("homeless"[All Fields] AND "persons"[All Fields]) OR "homeless persons"[All Fields] OR "homelessness"[All Fields]))) AND (("epidemiology"[Subheading] OR "epidemiology"[All Fields] OR "prevalence"[All Fields] OR "prevalence"[MeSH Terms]) OR magnitude [All Fields]).	2820

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Scopus	Depression* OR 'depressive symptom' OR 'depressive disorder' OR dysthymia* OR homeless* OR homelessness* OR prevalence* OR magnitude*	277
Web of Science	**TOPIC:** (depression OR depressive symptom OR depressive disorder OR dysthymia) *AND* **TOPIC:** (homeless OR homelessness) *AND* **TOPIC:** (prevalence OR magnitude) **Timespan:** All years. **Indexes:** SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI, CCR-EXPANDED, IC.	283

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	NA
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4-5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	7-8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	7-8
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	7-9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9-11
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	13

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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